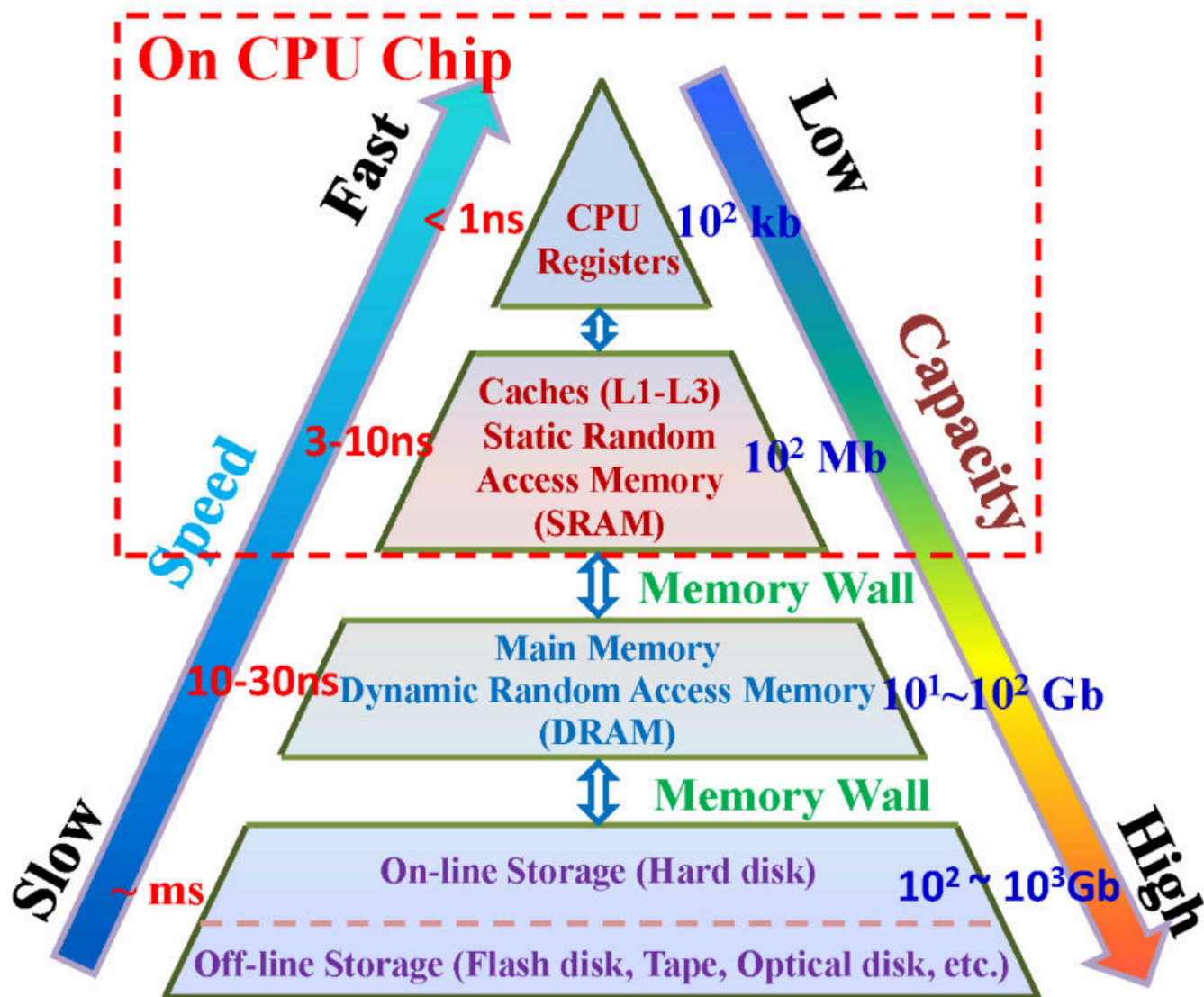


新型存储器的工作原理及发展趋势

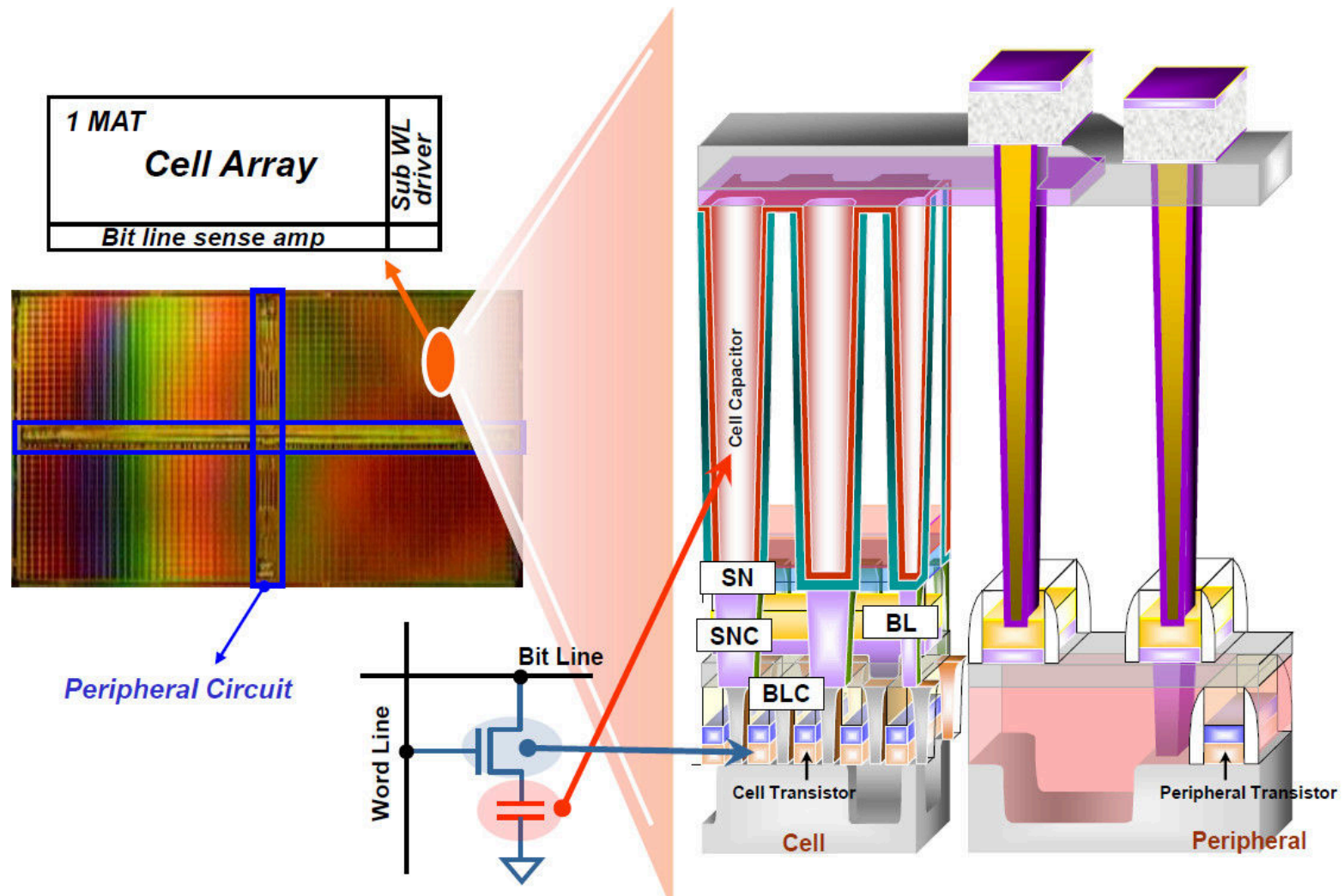
高滨

当前的存储器层级结构

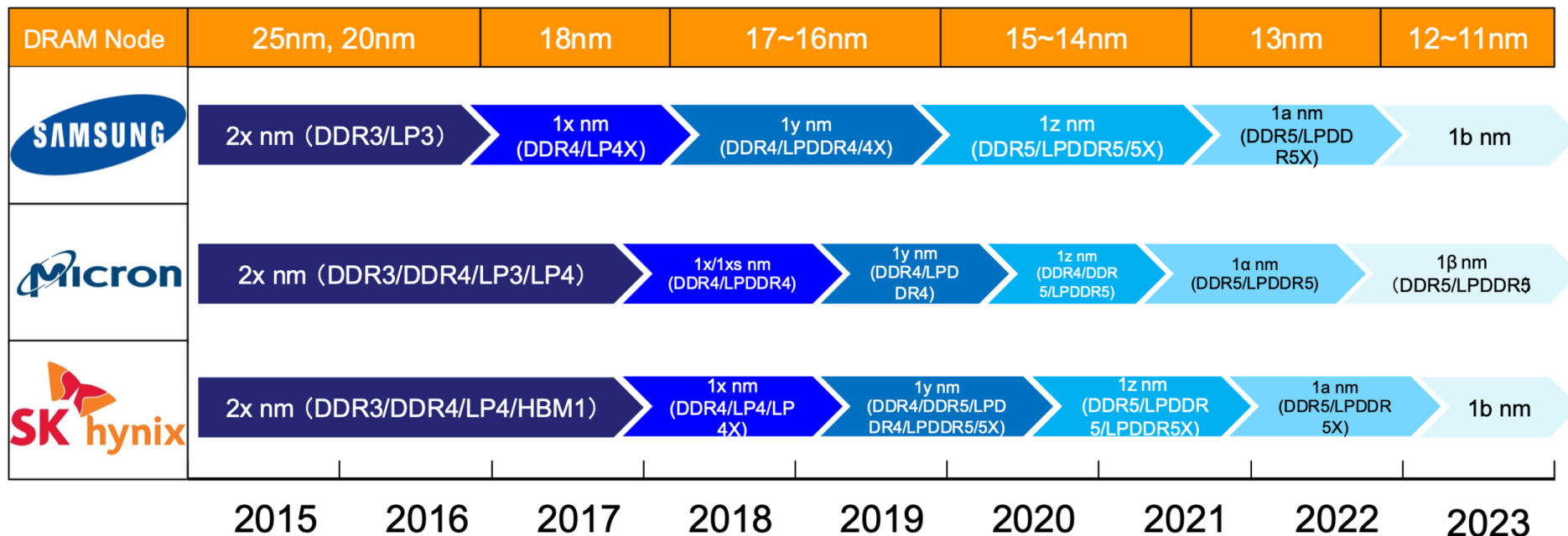


1. DRAM

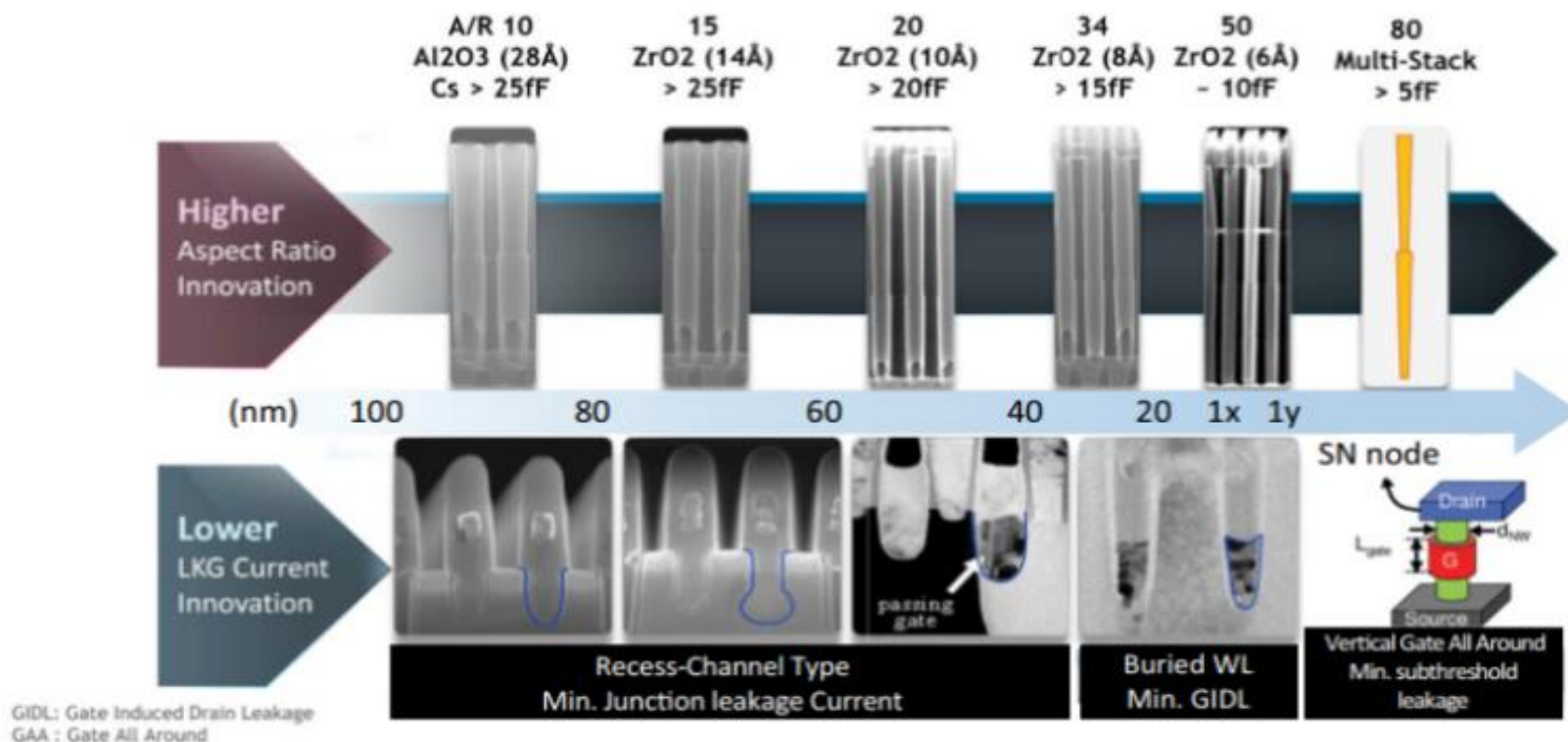
◆ 1个MOSFET串联1个电容 (1T-1C)



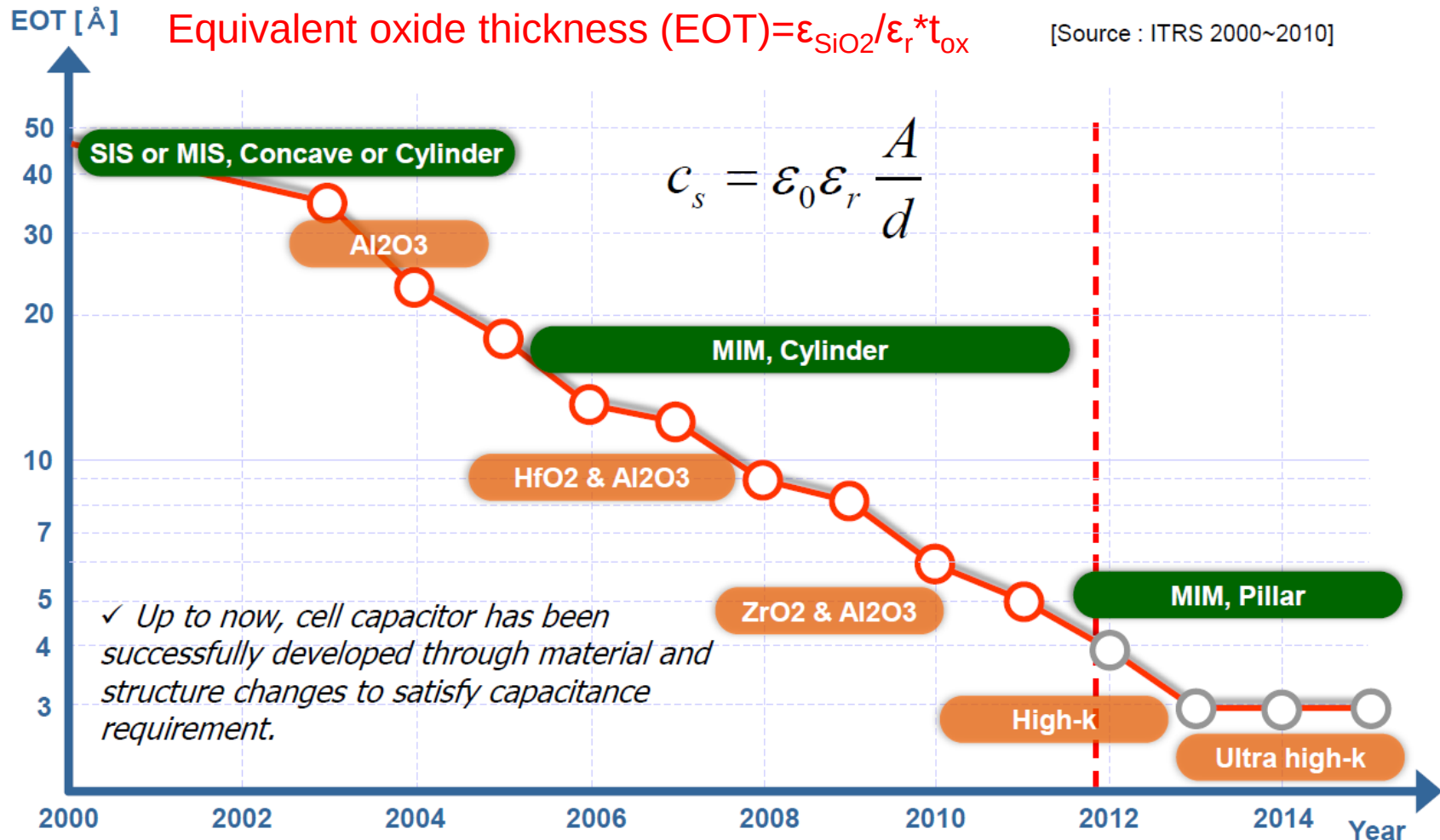
DRAM发展现状



推动DRAM持续微缩需要电容和晶体管的创新

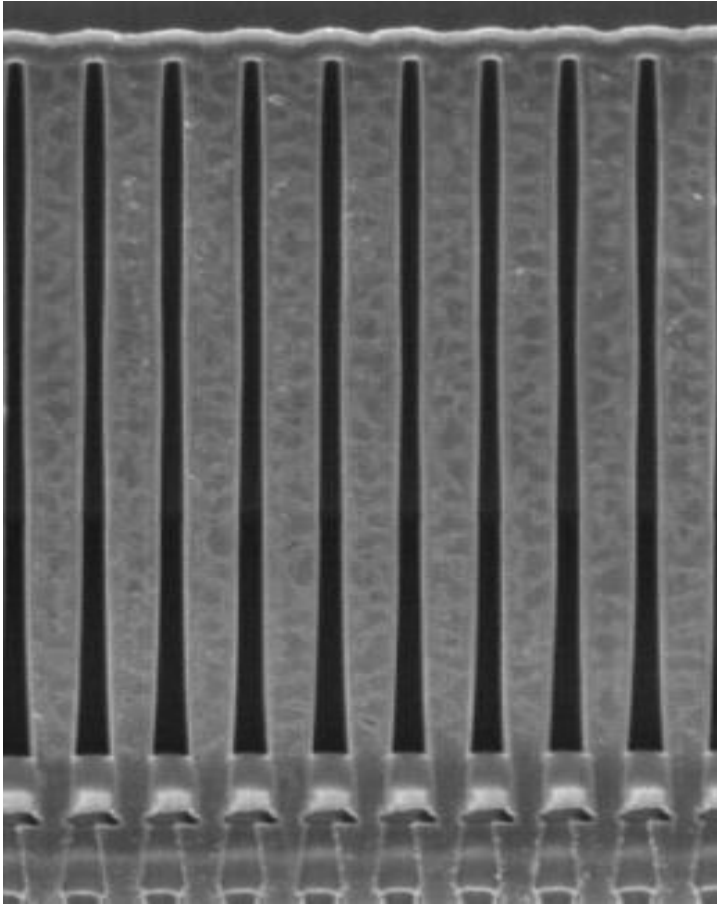


1.1 电容技术



Stacked Capacitor Formation

MIS



MIM

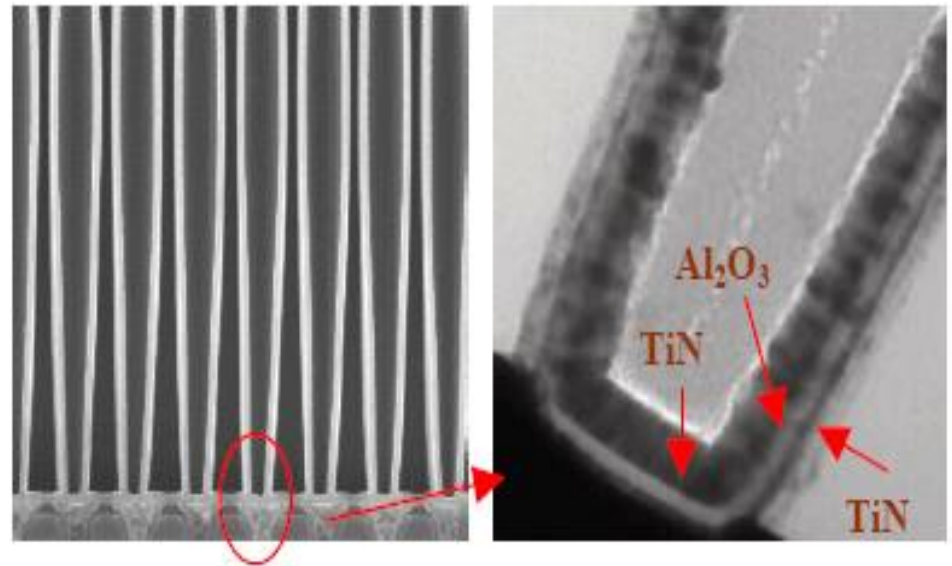
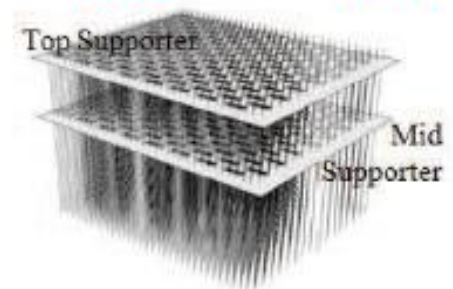
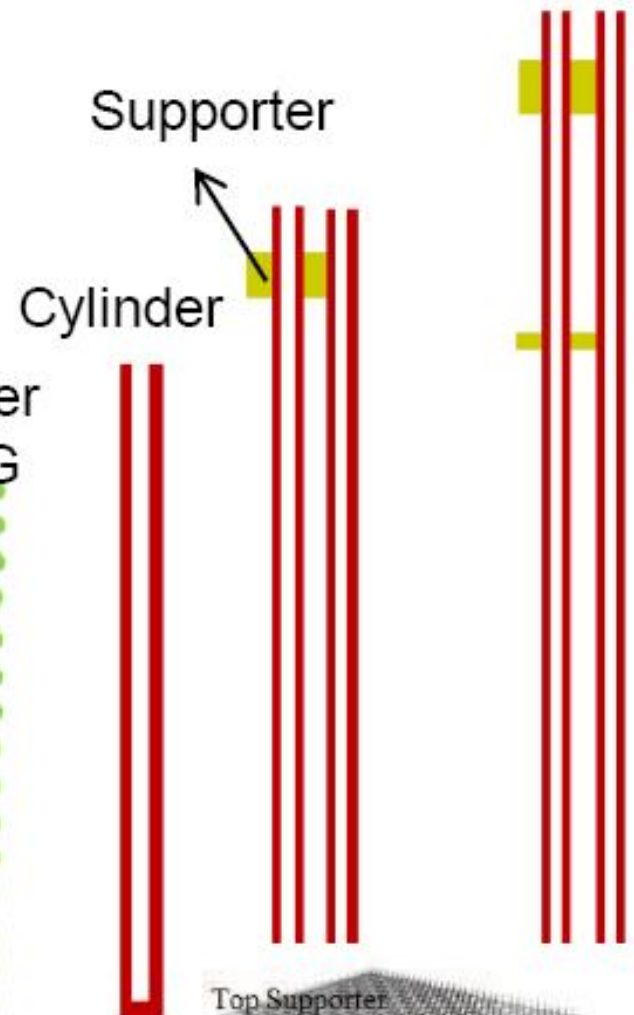
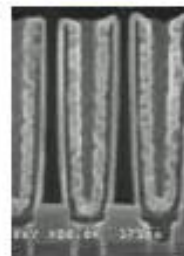
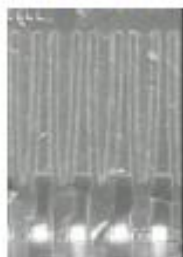
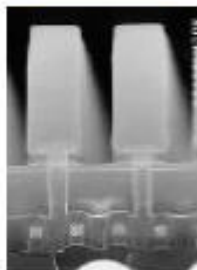


Fig.5. Cross-section of TiN/ Al_2O_3 /TiN MIM stack capacitor

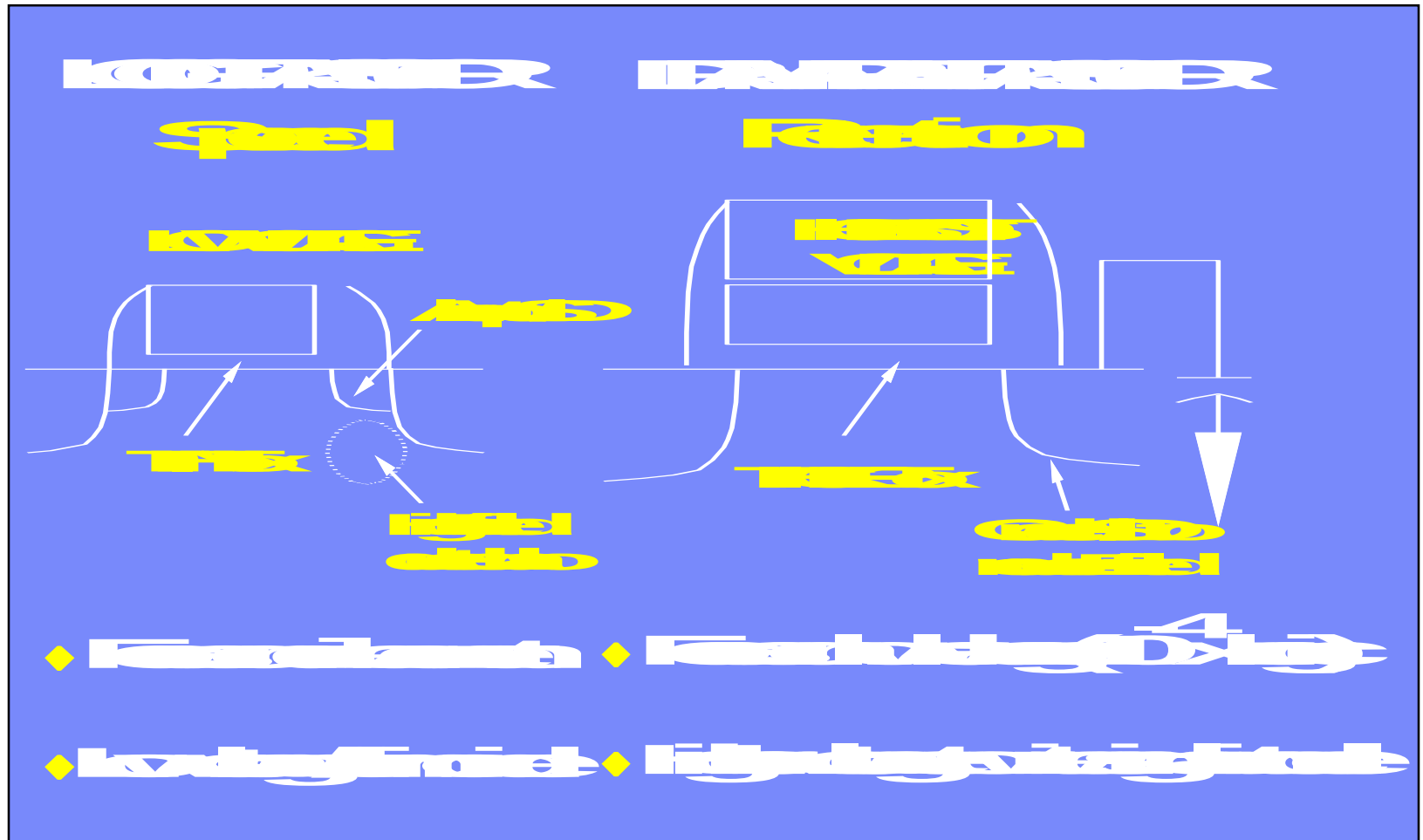
w/ HSG Poly bottom electrode

HSG : Hemispherical Grain
Hemispherical Silica Glass

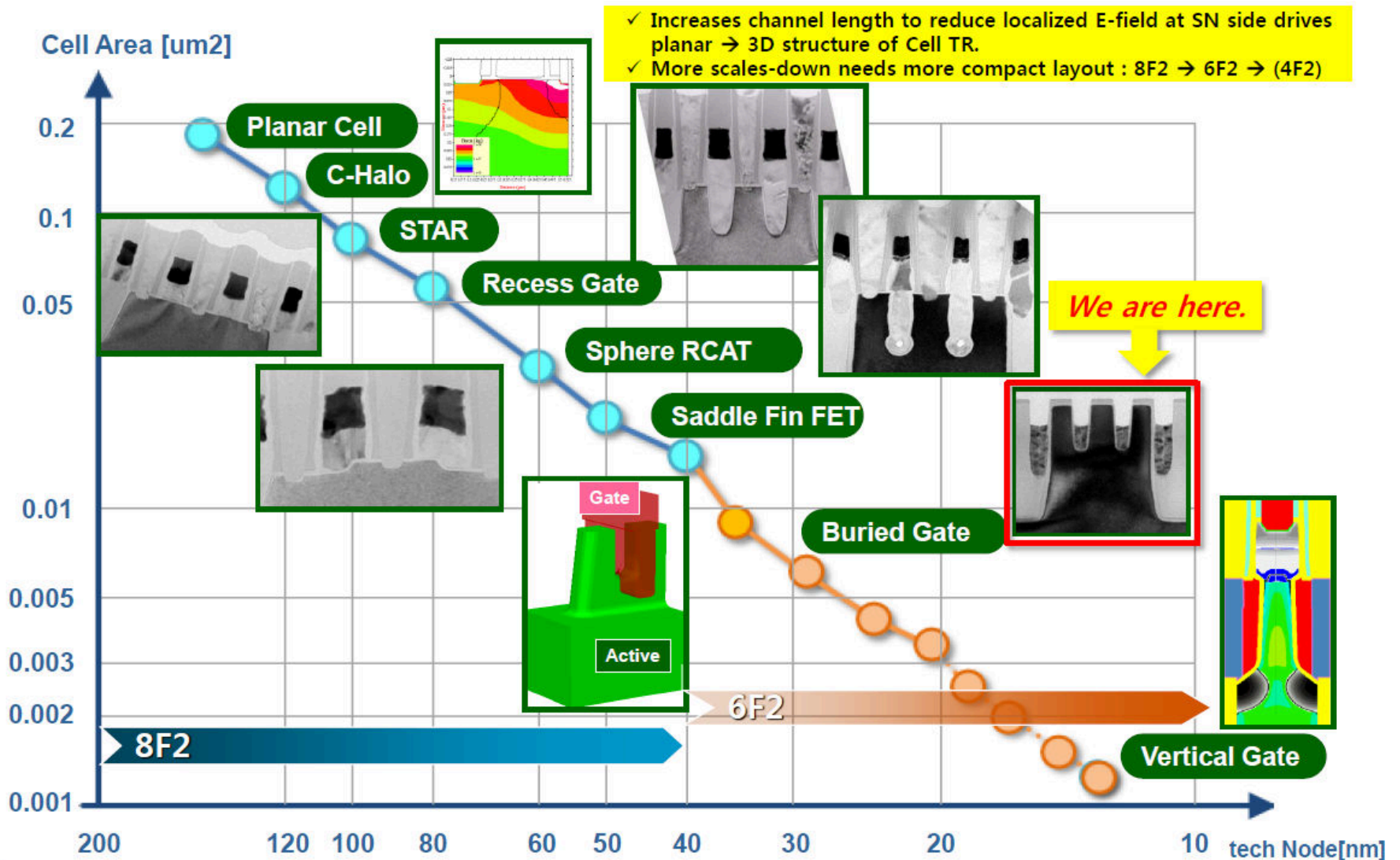


1.2 单元晶体管技术

Access Transistor

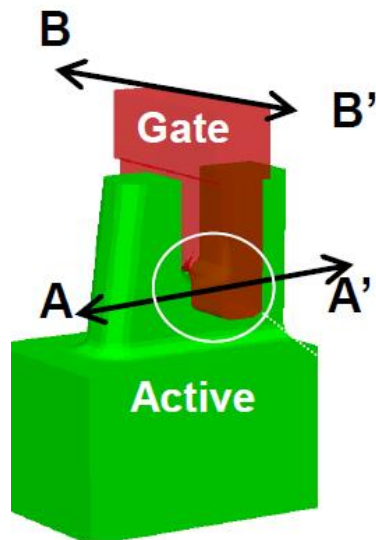


DRAM Scaling Roadmap (Access Transistor)

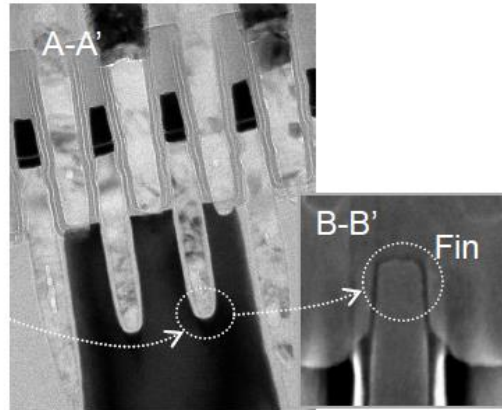


C.Y. Lee, IEDM 2015 short course

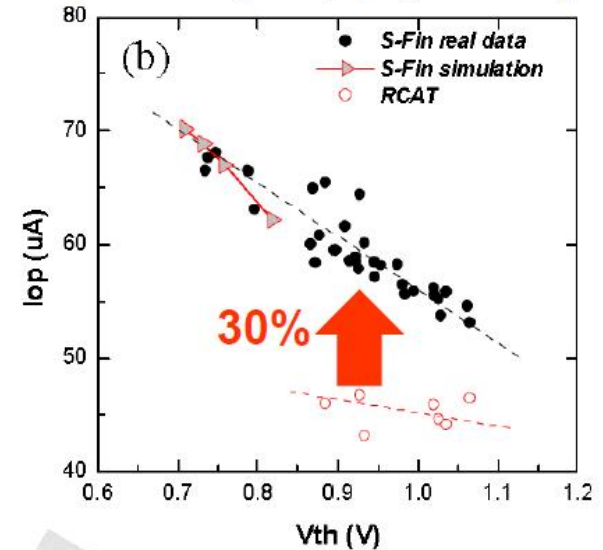
Saddle FinFET at 45 nm



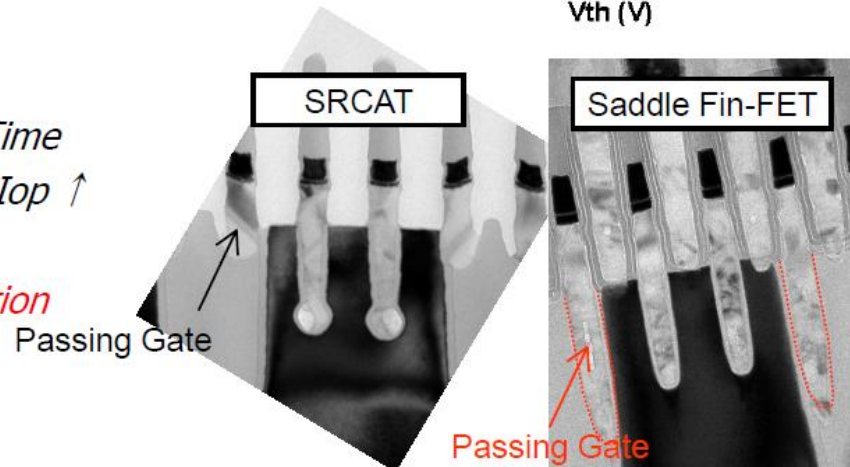
[H.J. Lee et al. (Hynix), VLSI 2008]



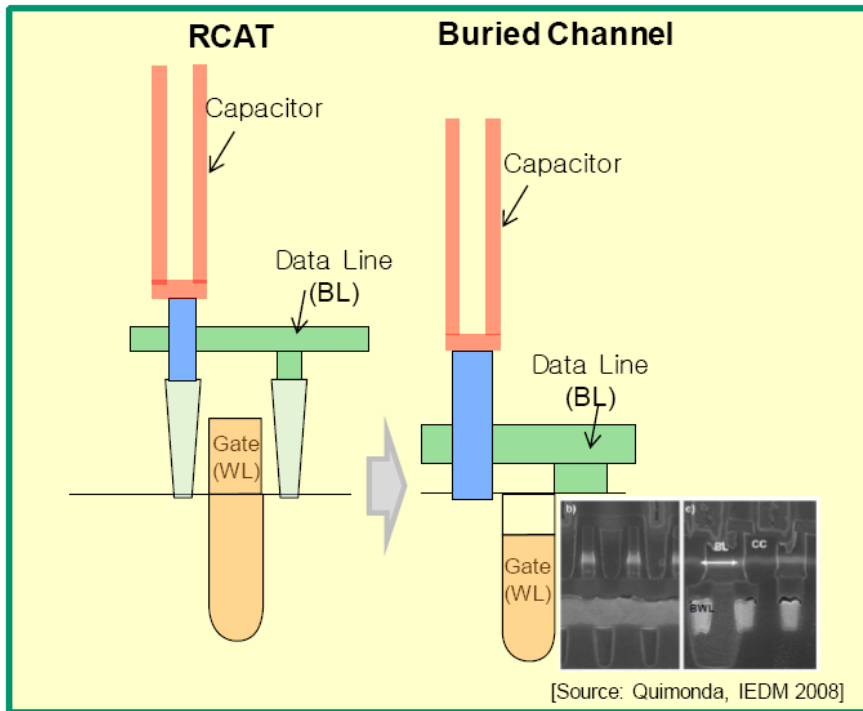
[S.-W. Chung et al. (Hynix), VLSI 2006]



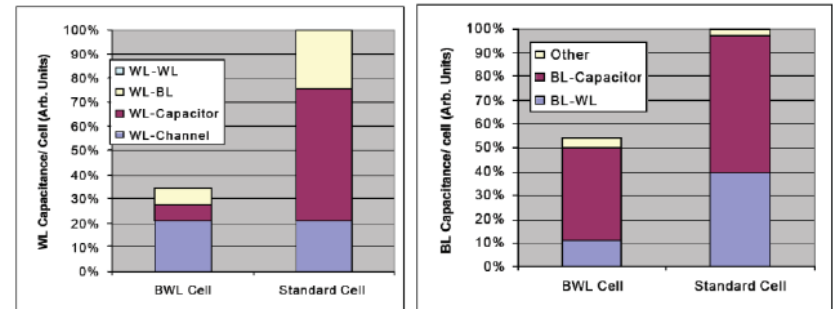
- ✓ *Saddle = Recess Gate + FinFET*
 - *Recess Gate: High Data Retention Time*
 - *Fin Gate: Body Effect ↓ / DIBL ↓ / Iop ↑*
- ✓ *Difficult to control the process variation*
- ✓ *GIDL by Passing Gate*



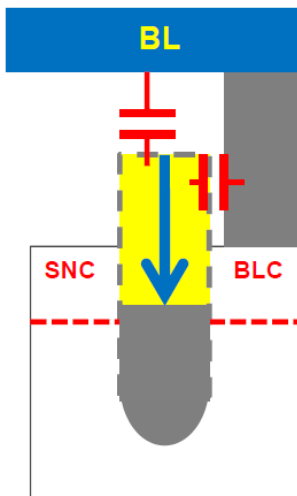
Buried Gate at 32 nm



- ✓ Reduction of WL related parasitic cap

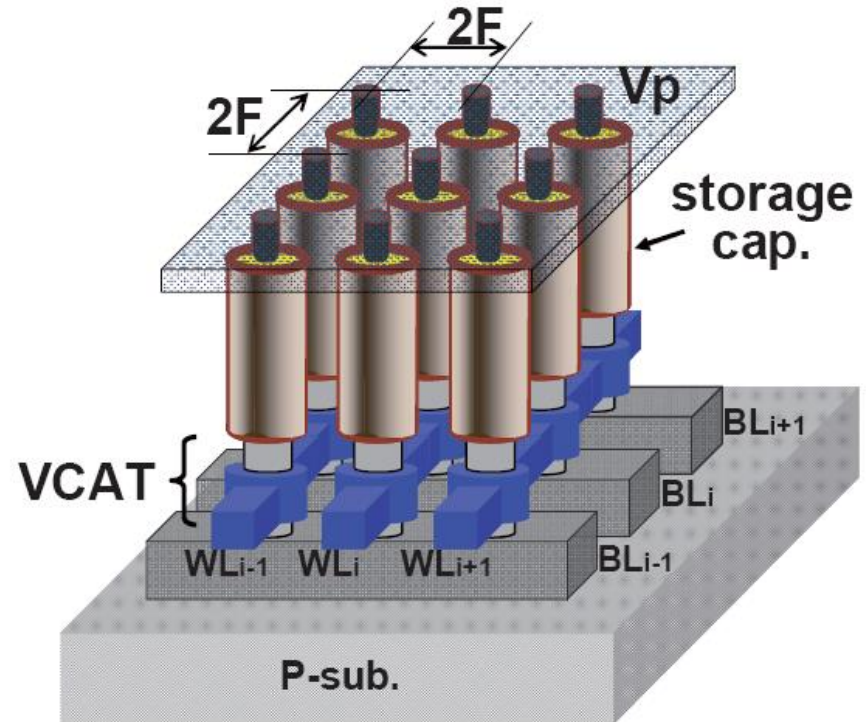
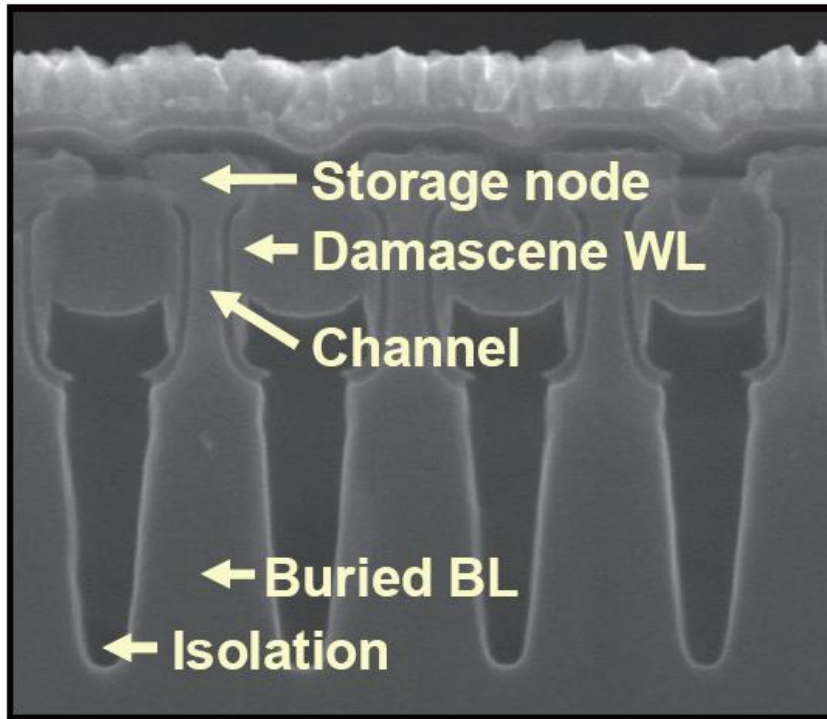


- ✓ More space for contact landing and fewer interface
- ✓ Small GIDL *E-field* due to WL metal rescess
- ✓ Large GIDL *E-field* due to metal work-function



- Reduced C_{bl} improves signal / sensing margin
- Still need to watch transistor leakage currents

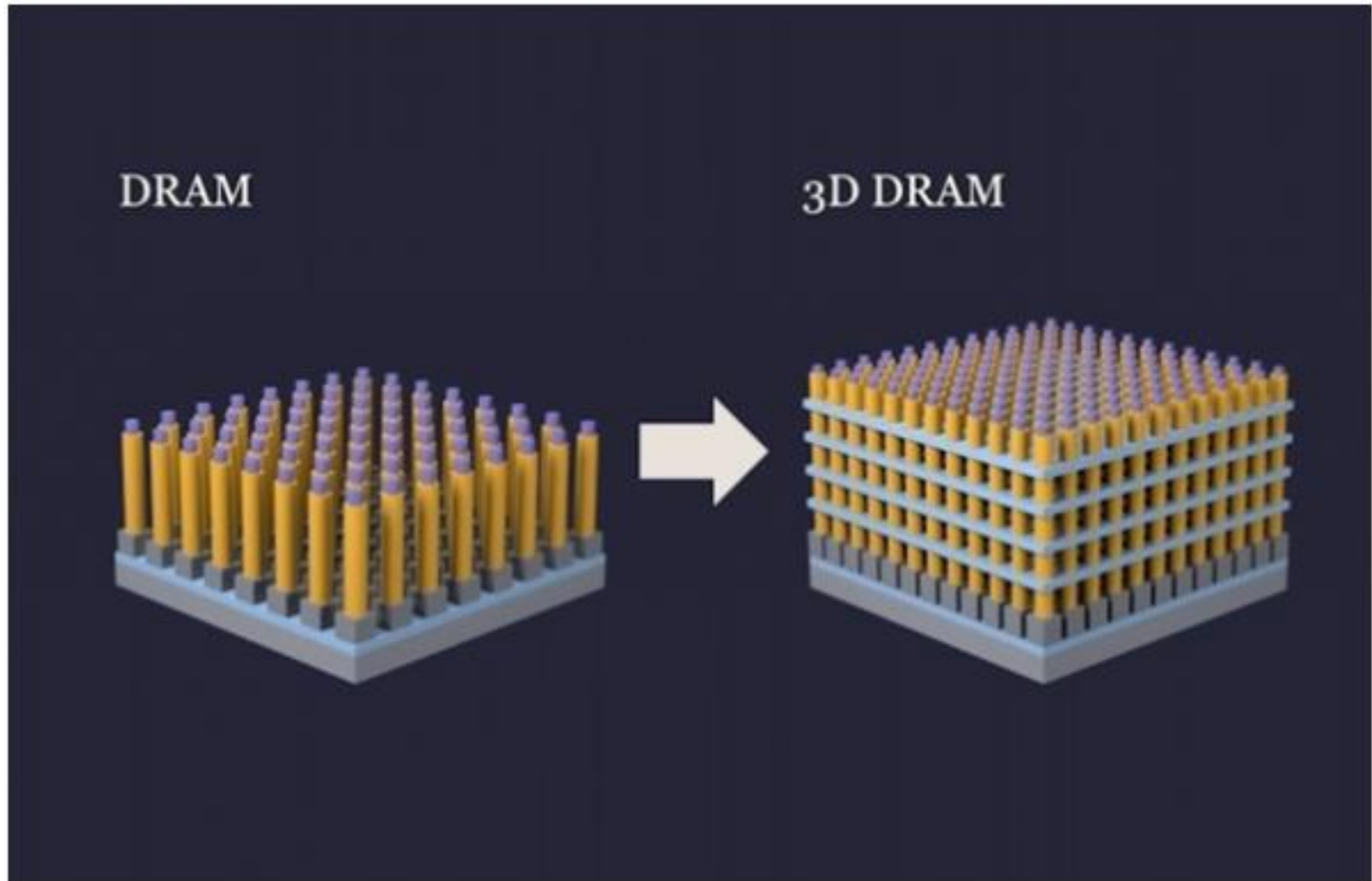
Vertical Transistor for 20nm beyond (Samsung)



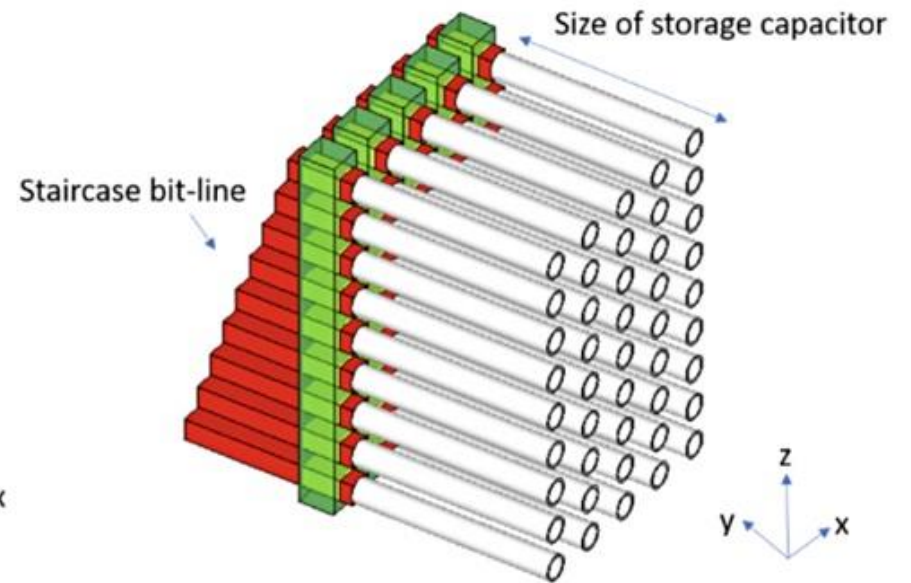
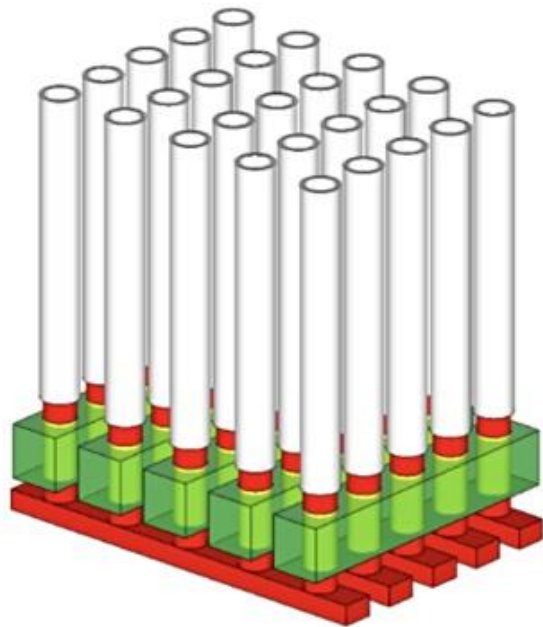
- Smallest possible cell size $\rightarrow 4F^2$
- Requires vertical access transistor
- **Not in volume production**, $6F^2$ with sub-half pitch active area used instead

Song et. al., VLSI Symp. 2009

1.3 3D DRAM

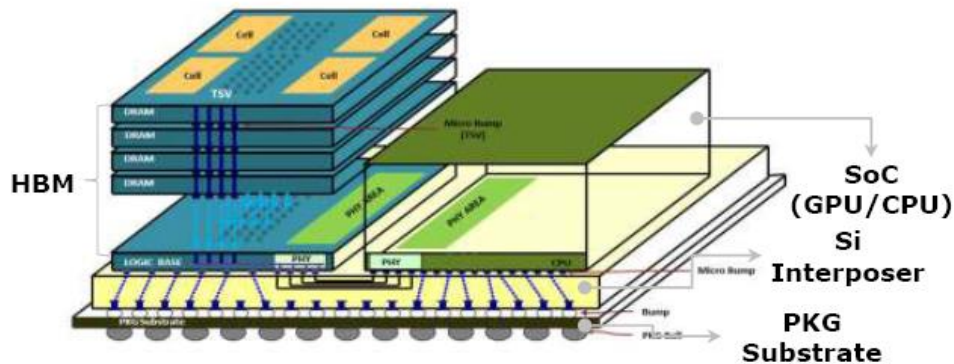


3D DRAM



1.4 HBM

Heterogeneous Chip Stack : HBM and SoC on Interposer



■ HBM to SoC Interconnection

- PHY through Si Interposer

■ SoC to Interposer, HBM to Interposer

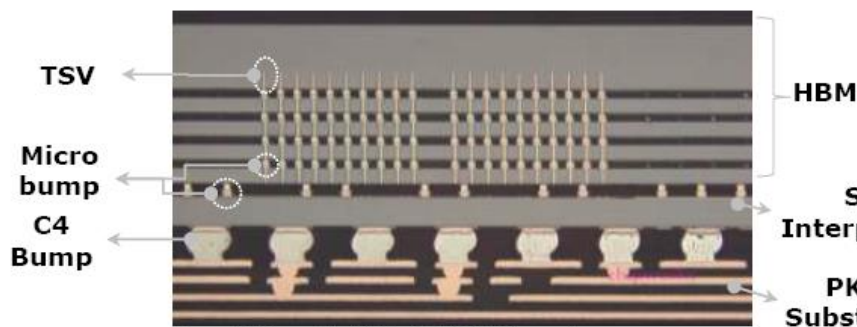
- Micro-Bump (Cu pillar bump)
- Same bump structure for SoC/HBM

■ Interposer to PKG Substrate

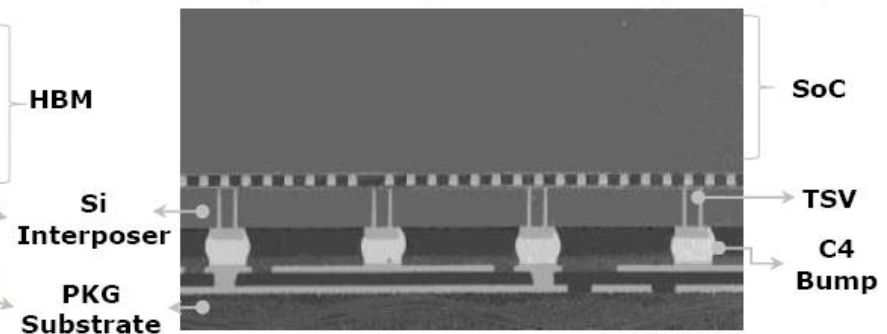
- C4 Bump

■ HBM Inside

- Logic Buffer (Base Die) + Multiple DRAM (Core Die)
- TSV/Micro-bump Interconnect (>1000ea per chip)



Source) Chipworks (SK Hynix's HBM1 Image)

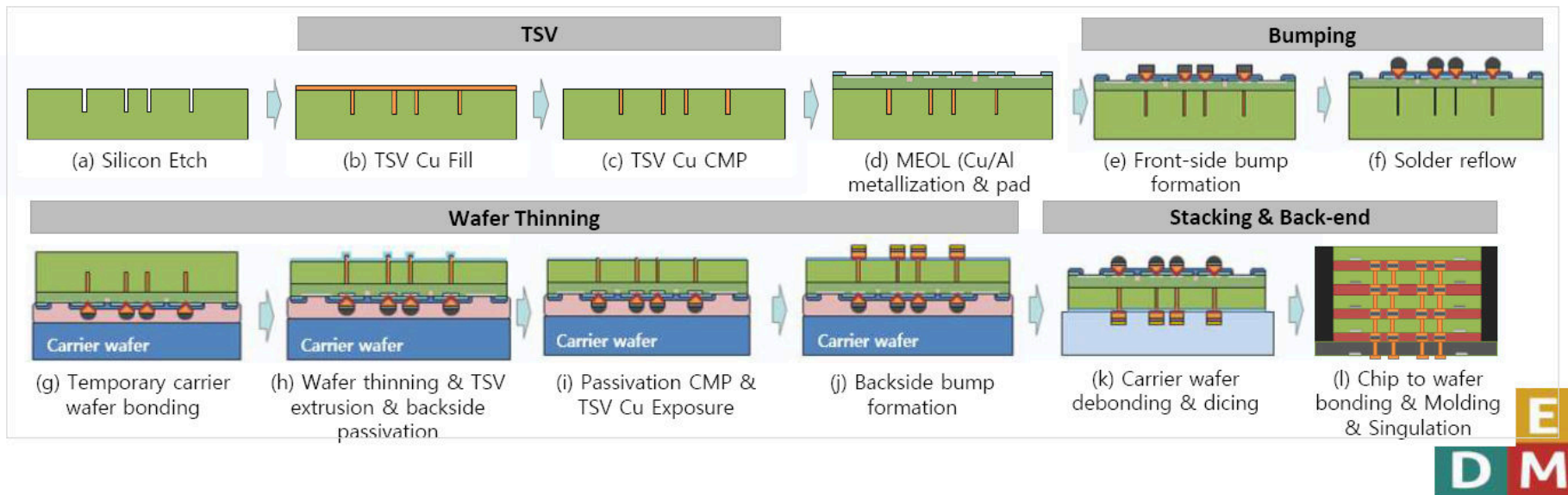


Source) SK Hynix

HBM Overall Process Flow

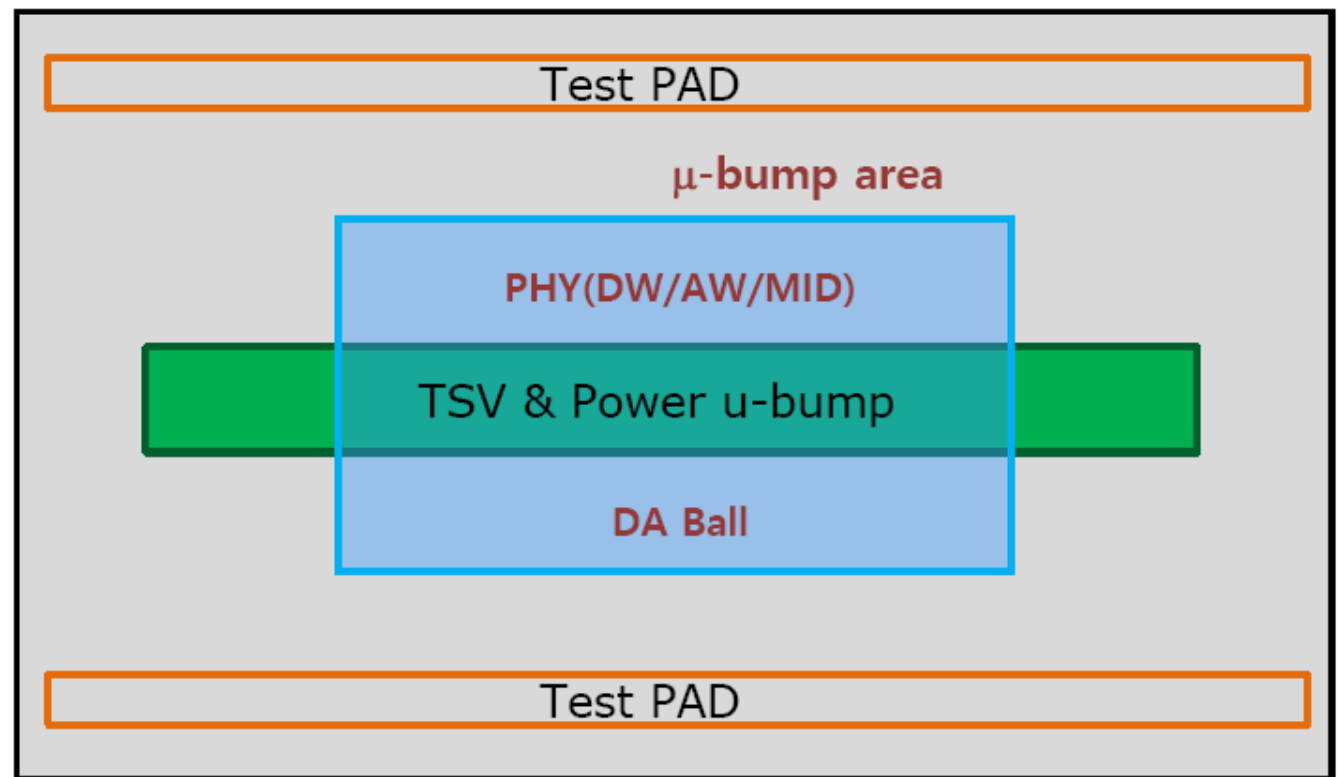
Via Middle TSV + Double Side Bumping + Chip to Wafer Stacking

FEOL	TSV	MEOL	Wafer Test	Bumping	Wafer Thinning	Stacking	Back-End
- CMOS Process - DRAM Capacitor (Core Die Only)	- Via Middle	- Cu/Al Metallization	- KGD (Known-Good Die) Wafer	- Front/Back-side bumping	- Thin Wafer Handling - Backside Via Reveal	- C2W Stack	- Mold, Mold Grind, De-bond - Stacked Wafer Test



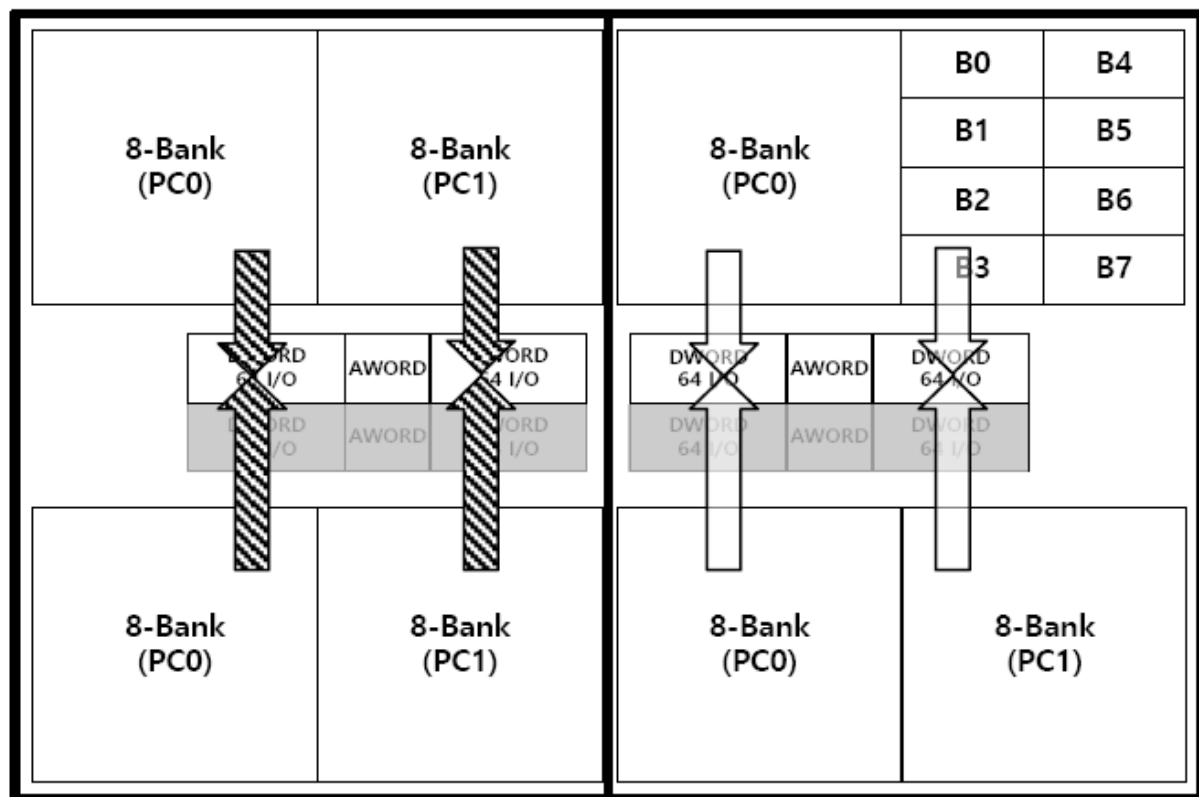
HBM Architecture – Buffer die (1)

- Buffer die architecture
 - Routing from Core dies to external interface
 - Testability for mass production and failure analysis

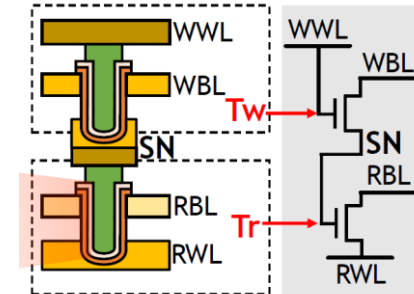
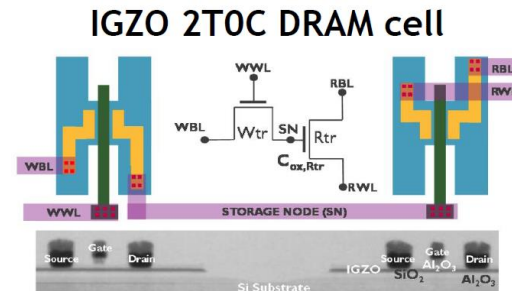
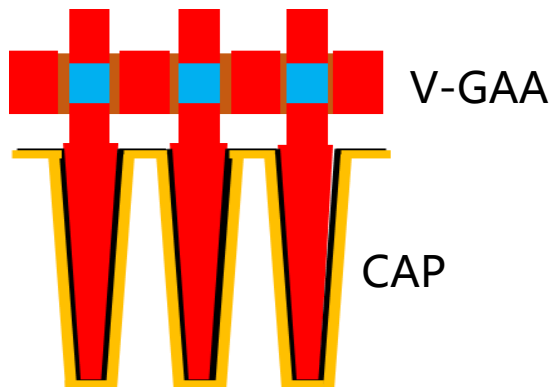


HBM Architecture – Core die

- Core die is a DRAM die for HBM
 - 2 channels per die, 2 pseudo channels per 1channel
 - 16 banks: 4 bank groups, 4 banks per 1bank group
 - Total 64 banks per each chip

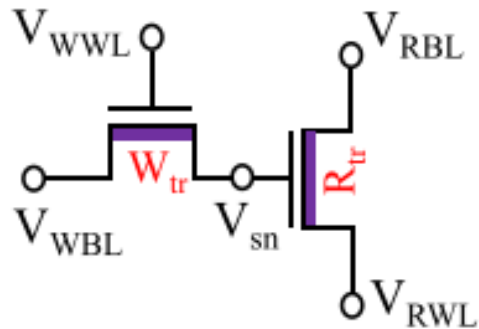
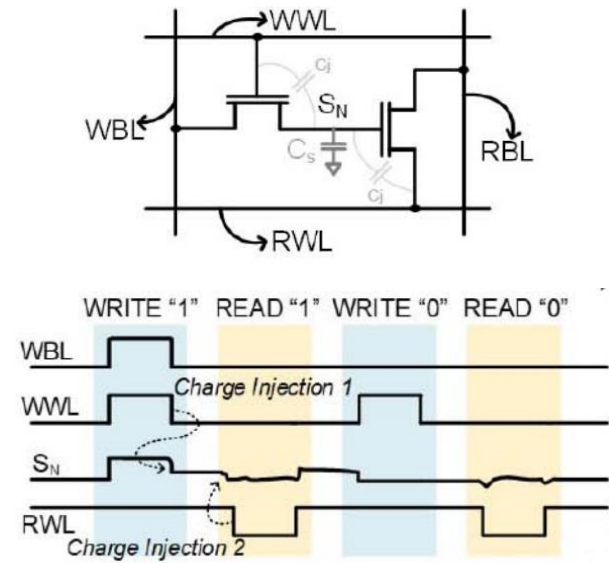
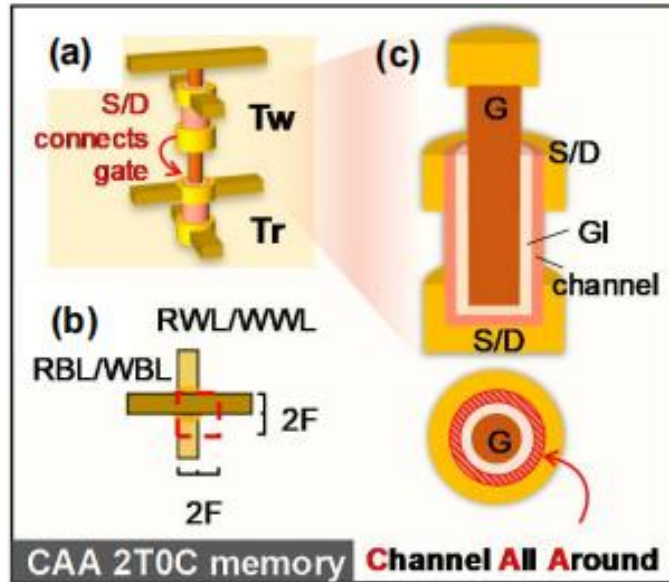


1.5 eDRAM



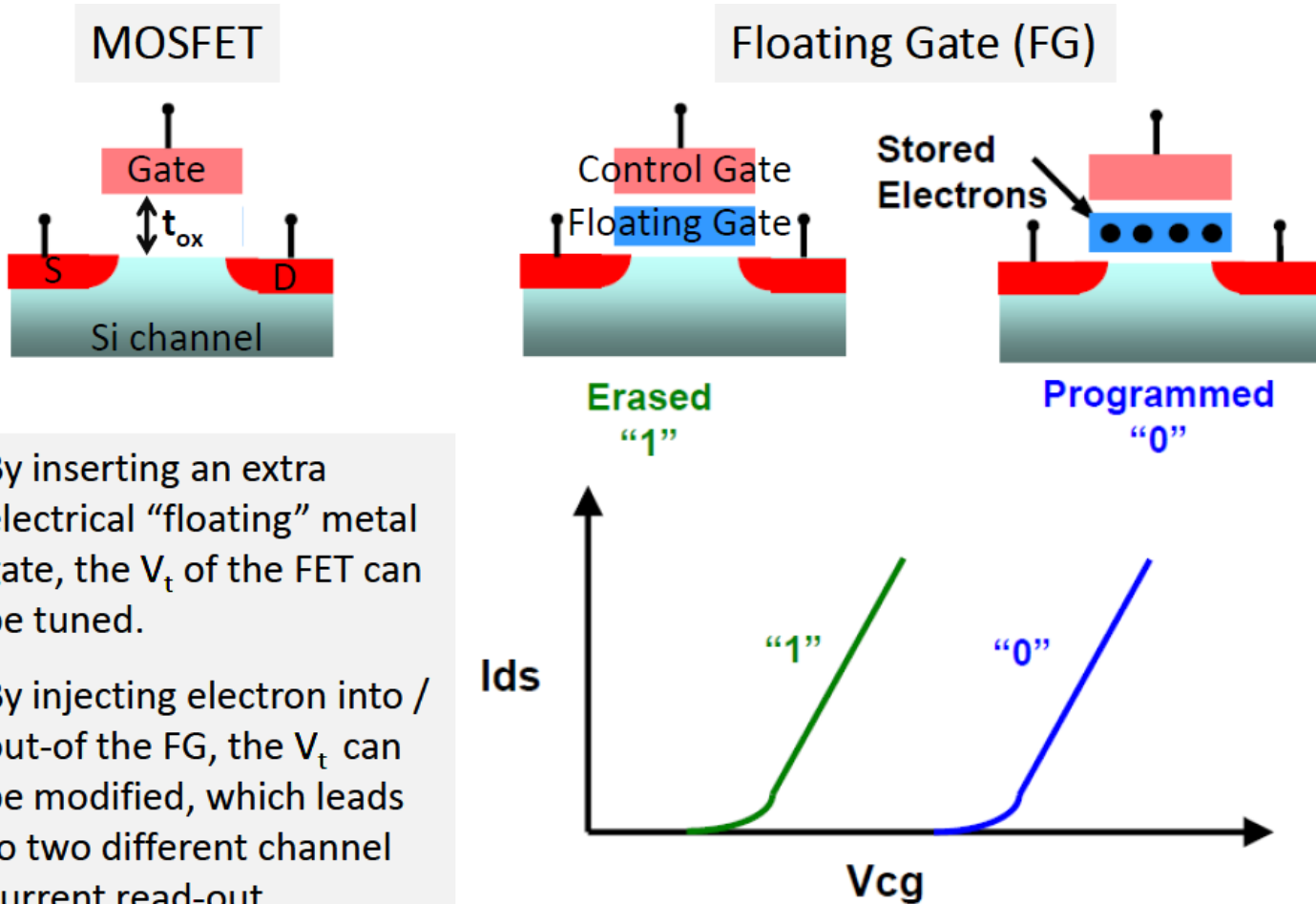
- **BEOL transistors**
 - Poly-Si channel, oxide channel TFT (e.g. IGZO, ITO, IWO)
 - Planer transistor, vertical transistor
- **On-chip high-density capacitor**
 - BEOL high-k dielectric
 - 2T0C or 1T0C

2T0C cell structure



2. 3D NAND

- 快闪存储器Flash，分成NOR Flash和NAND Flash两种
- 特殊设计的晶体管结构



- By inserting an extra electrical "floating" metal gate, the V_t of the FET can be tuned.
- By injecting electron into / out-of the FG, the V_t can be modified, which leads to two different channel current read-out.

(Sources: P. Kalavade, intel)

3D NAND发展现状



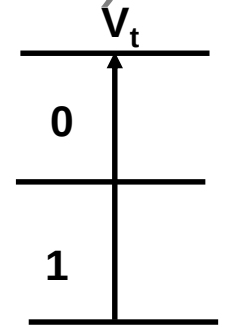
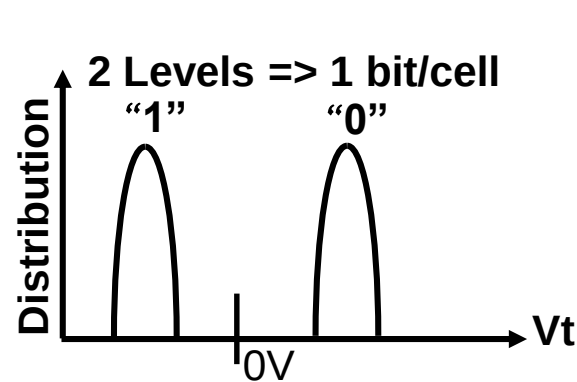
2.1 多比特存储

Single Bit Cell (SBC/SLC) & Multi Level Cell (MLC)

1 bit/cell: 2 states:

Level 0 = Erased

Level 1 = Programmed



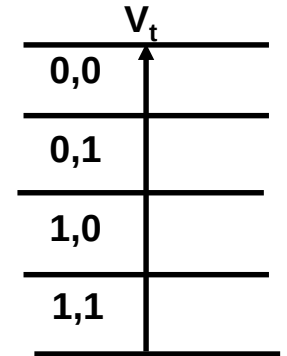
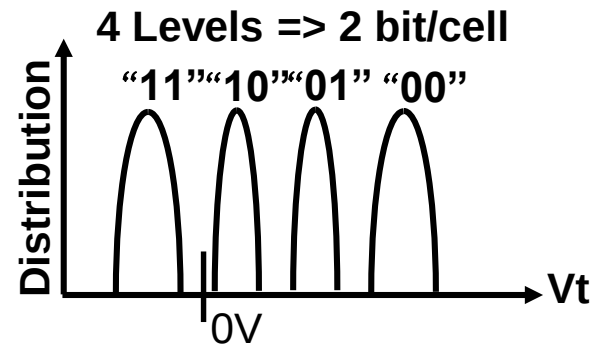
2 bits/cell: 4 states:

Level 0 = Erased

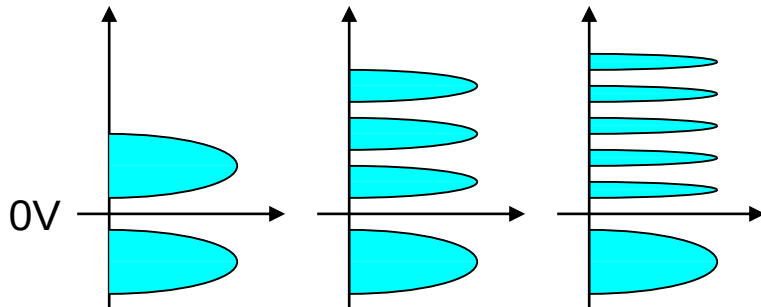
Level 1 = Programmed to L1

Level 2 = Programmed to L2

Level 3 = Programmed to L3



- Having the threshold voltage of the cells defined to 2^n different levels allows for storing n bits per cells



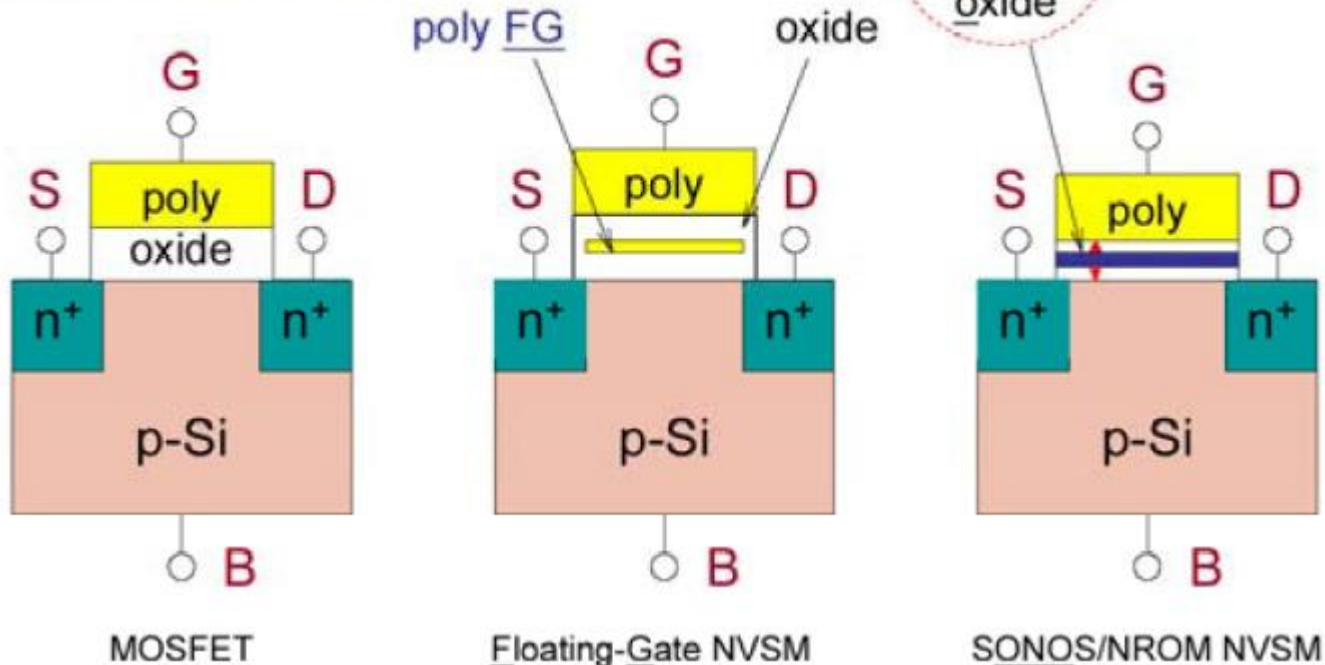
Single Bit/cell $\rightarrow$ MLC $\rightarrow$ N-bit/cell,
 $N = \log(\text{\#levels}) / \log(2)$

2.2 电荷俘获型单元 (CTM)

Charge-Trap FLASH-SONOS

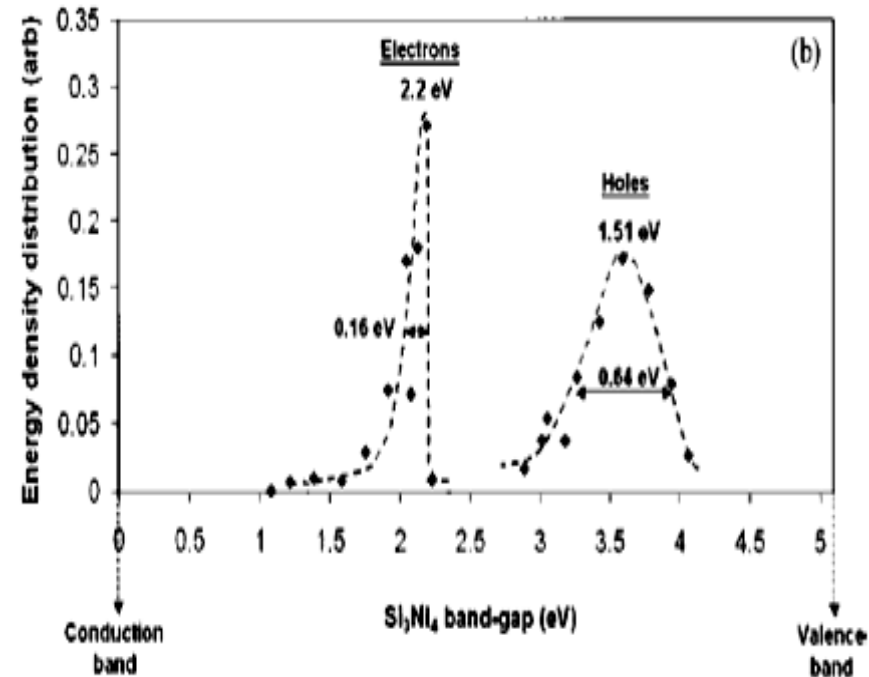
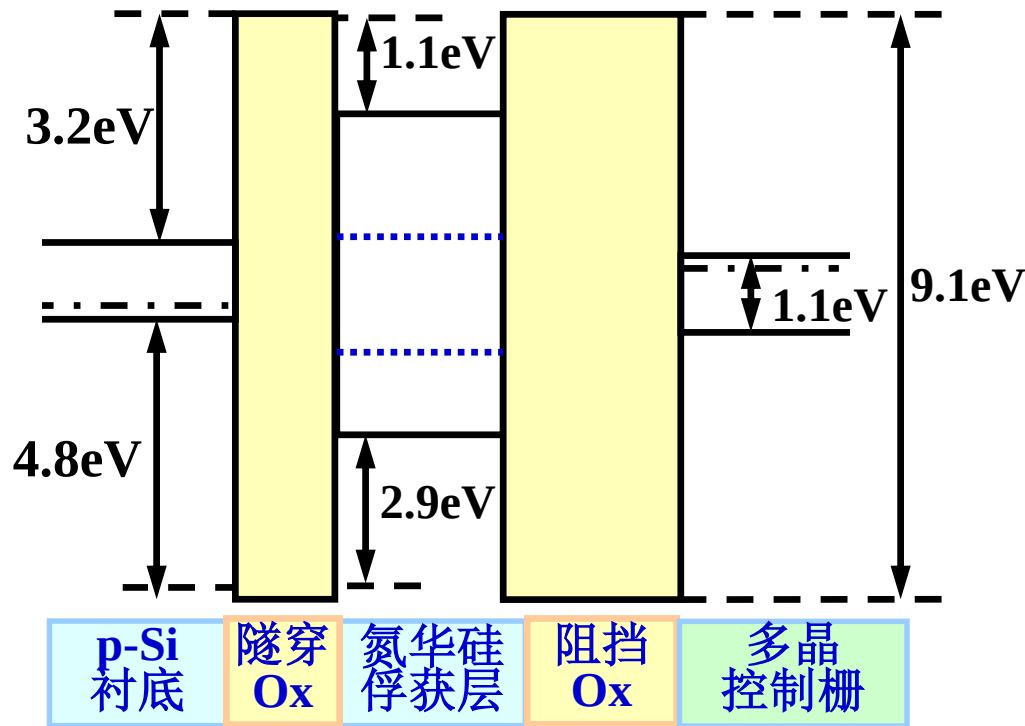
- **Floating gate**
- **Charge trap (in dielectric materials) – SONOS**

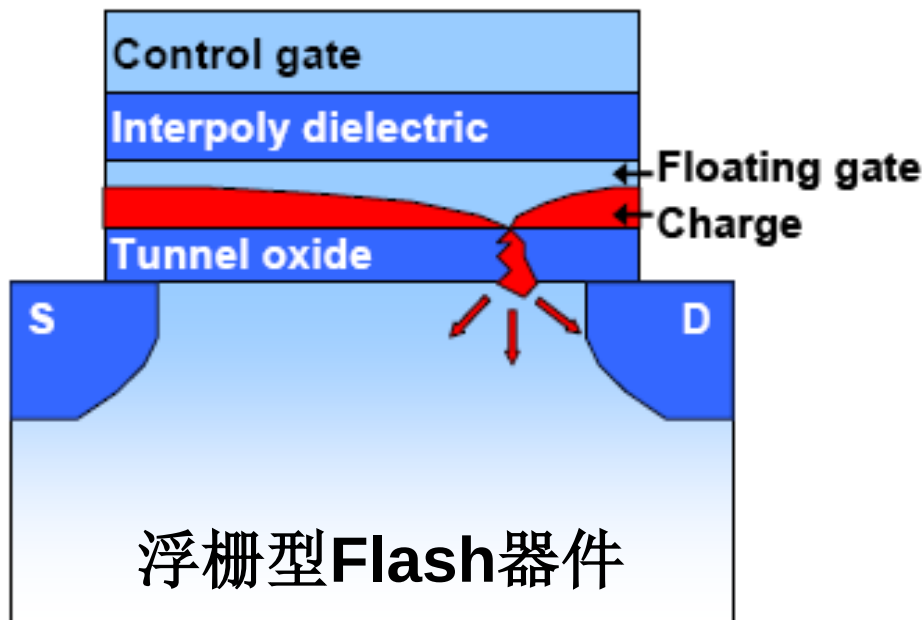
A special MOSFET with a charge storage layer



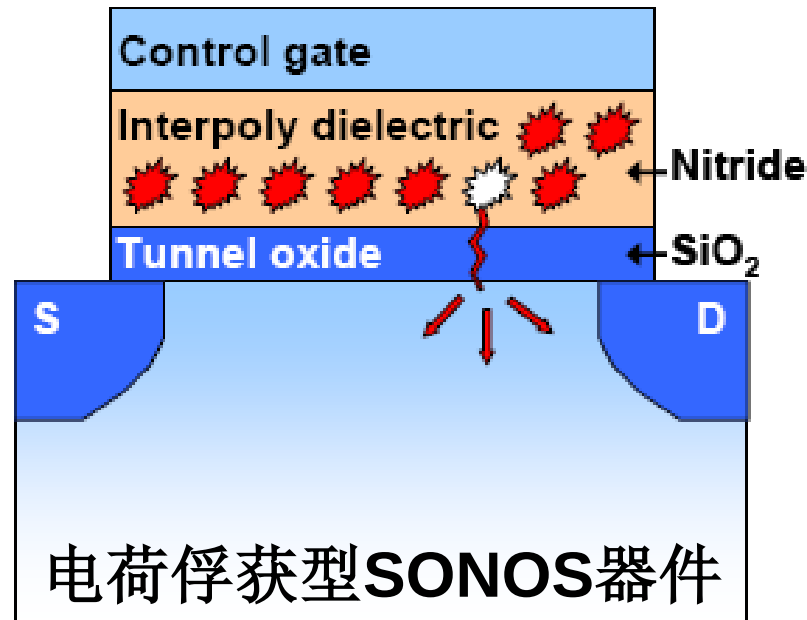
SONOS存储器件中的俘获材料及其制备

- LPCVD Si_3N_4 (SiH_4 and NH_3)
- Si_3N_4 has trap density $\sim 10^{19} \text{ cm}^{-3}$
- Trap energy is deep into the nitride bandgap
- Traps can store positive or negative charges



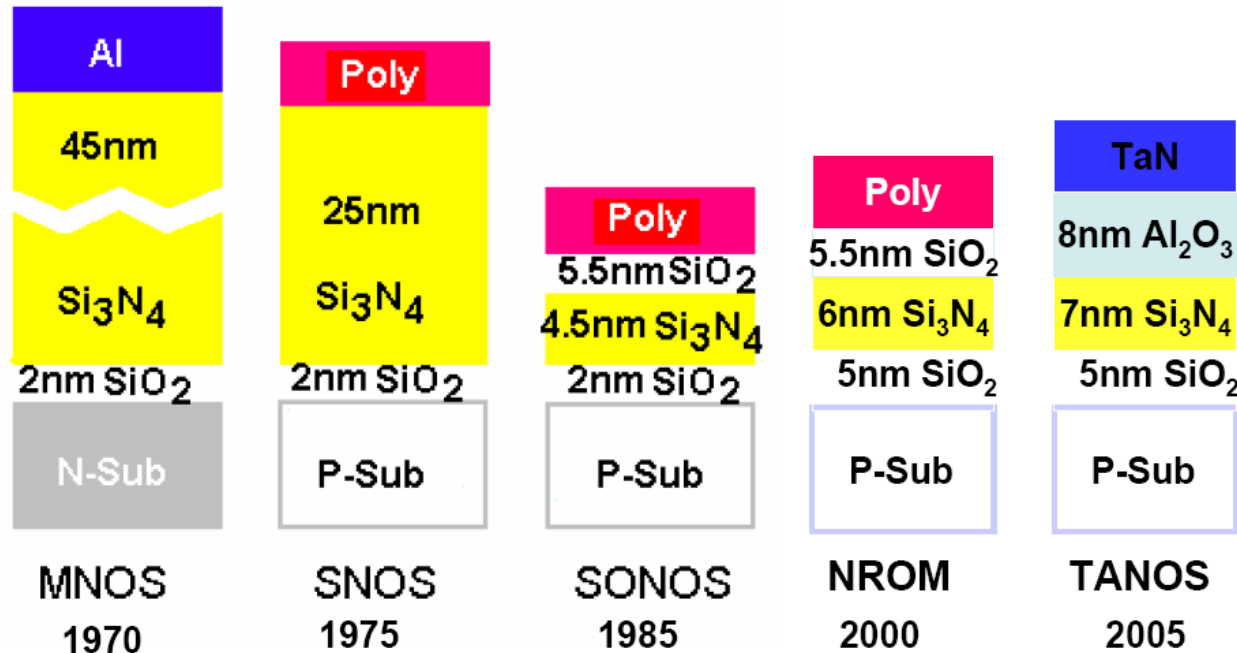


- Poly浮栅中电荷可以自由移动，单一缺陷就可以导致电荷流失；
- 相邻元件存在浮栅间电容耦合干扰
- 多级存储需要控制存储电荷数量；
- 栅结构复杂



- 氮化物存储层中电荷被存在电荷阱中，电子无法自由移动（retention和endurance提升）；
- 相邻元件不存在耦合干扰；
- 可实现多物理位存储；
- 栅结构简单，利于工艺集成
- SONOS的操作电压更低（<10V, typically 5~7V）

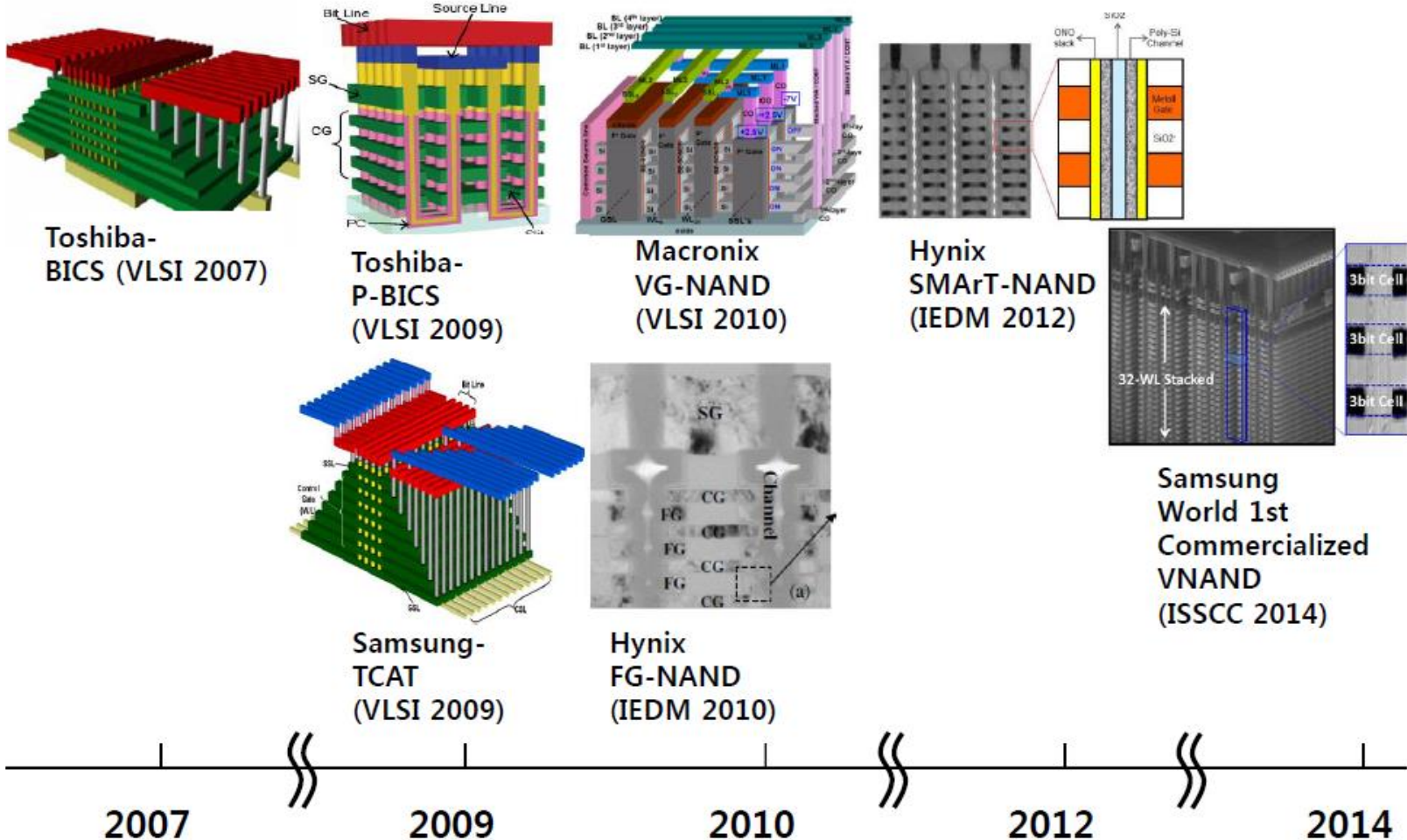
Evolution of Nitride-Based Charge-Trap FLASH



- Driving force is low voltage, low power and high density
- ONO and ANO structure allows scaling of the nitride and blocks charge injection from/to the gate electrode
- Scaled voltage with improved retention
- High-K Al₂O₃ oxides and Metal Gate electrodes offer new possibilities for improved NVM performance

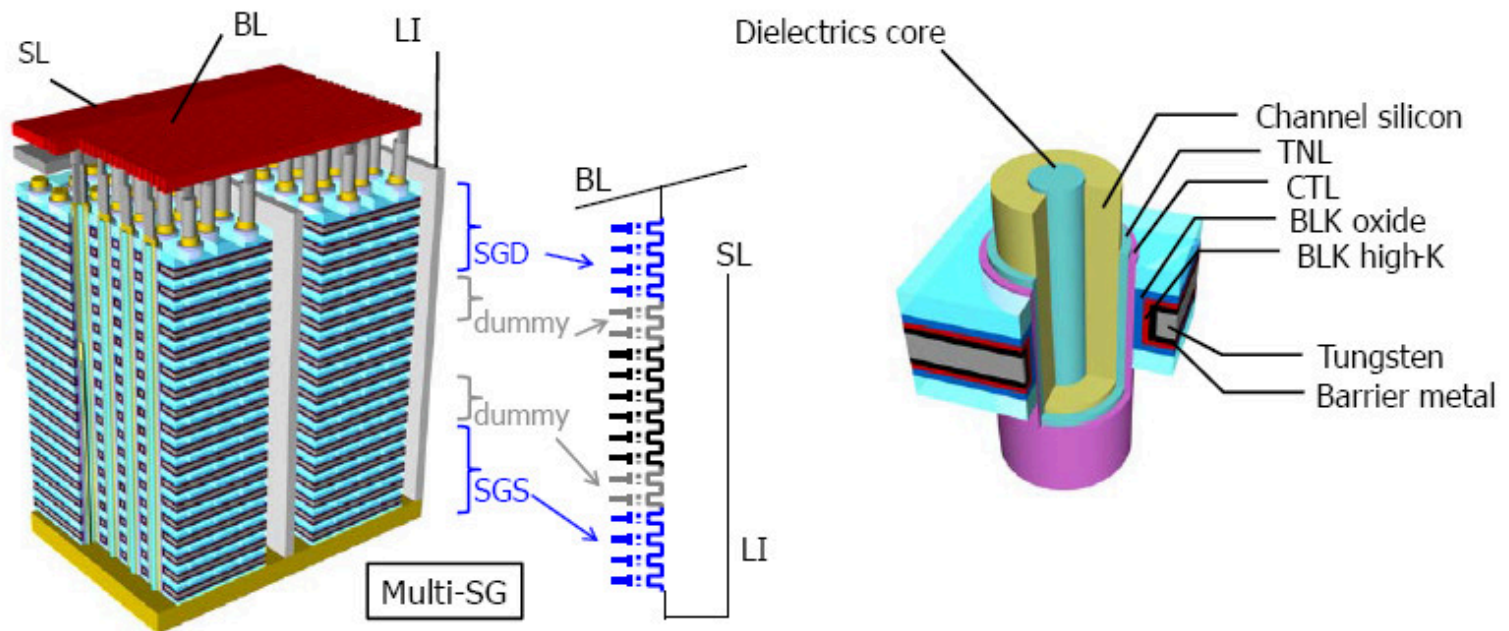
2.3 3D NAND

3D NAND R&D



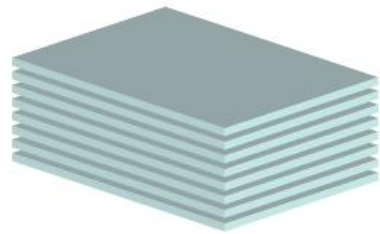
Y. Park, IEDM 2015 short course

3D NAND Architecture

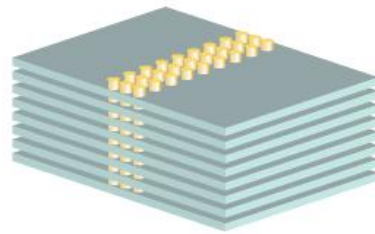


- Cylinder cell, shielding bit line to bit line coupling
- 3D block consists of multiple 2D blocks
- Multiple strings connected to 1BL (Only 1 string to 1BL in 2D NAND)
- Ever larger block sizes (32KB in 0.16um 1Gbit in 2000, >18MB in 96L in 2018)

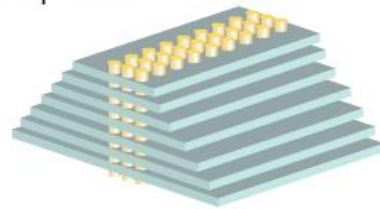
Process Flow of Gate-Last Fabrication



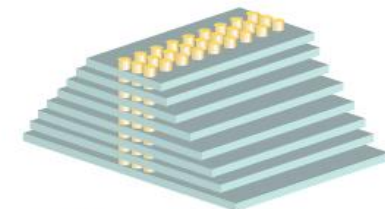
(a) Oxide/Nitride Multi-Layer deposition



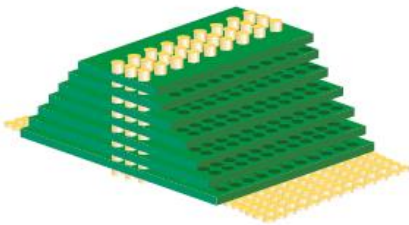
(b) Channel Hole deposition



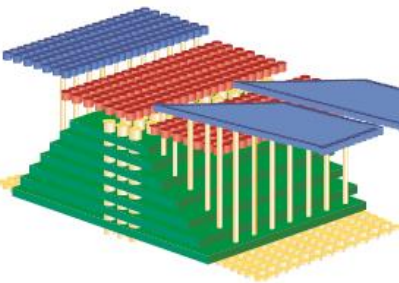
(c) Gate pad



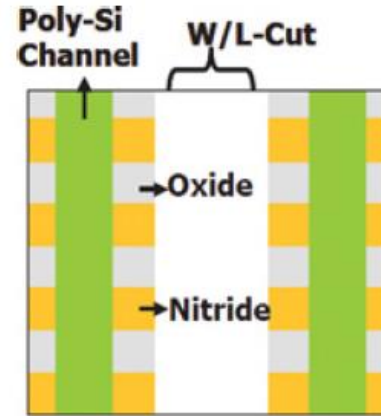
(d) W/L Cut Etch



(e) After Gate Replacement Process & CSL Implant



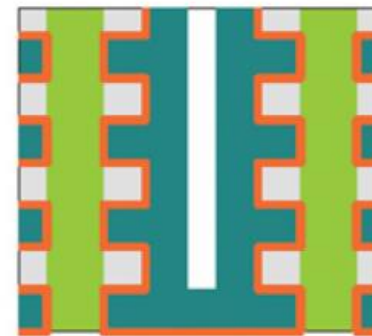
(b) BEOL



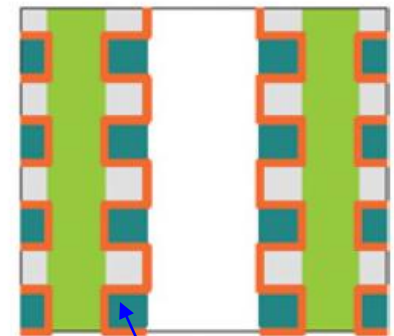
(a) After 'W/L cut' dry etch



(b) Wet removal of nitride



(c) Deposition of gate dielectric and tungsten

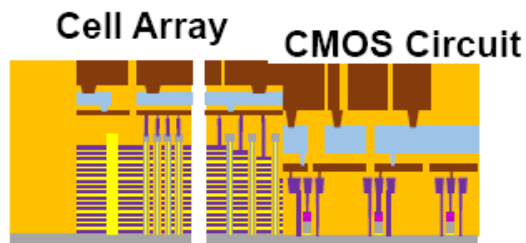


(d) Gate node separation

W-gate reduces WL resistance

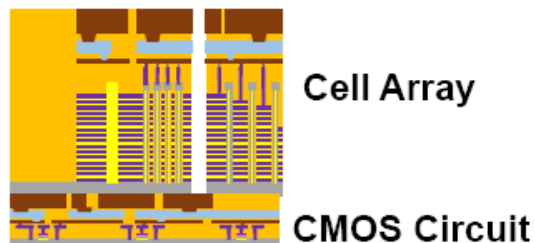
J. Jang et al., "Vertical Cell Array using TCAT...", Symp. VLSI Tech. paper 10A-4, p. 192 (2009)

Architecture Choices: Where to place the CMOS



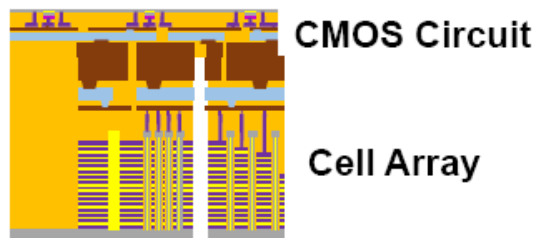
CMOS Next to Array
aka: next door parking

Lowest process/CapEx cost.
Simple, traditional.
But larger die area, and less
parallelism in program.



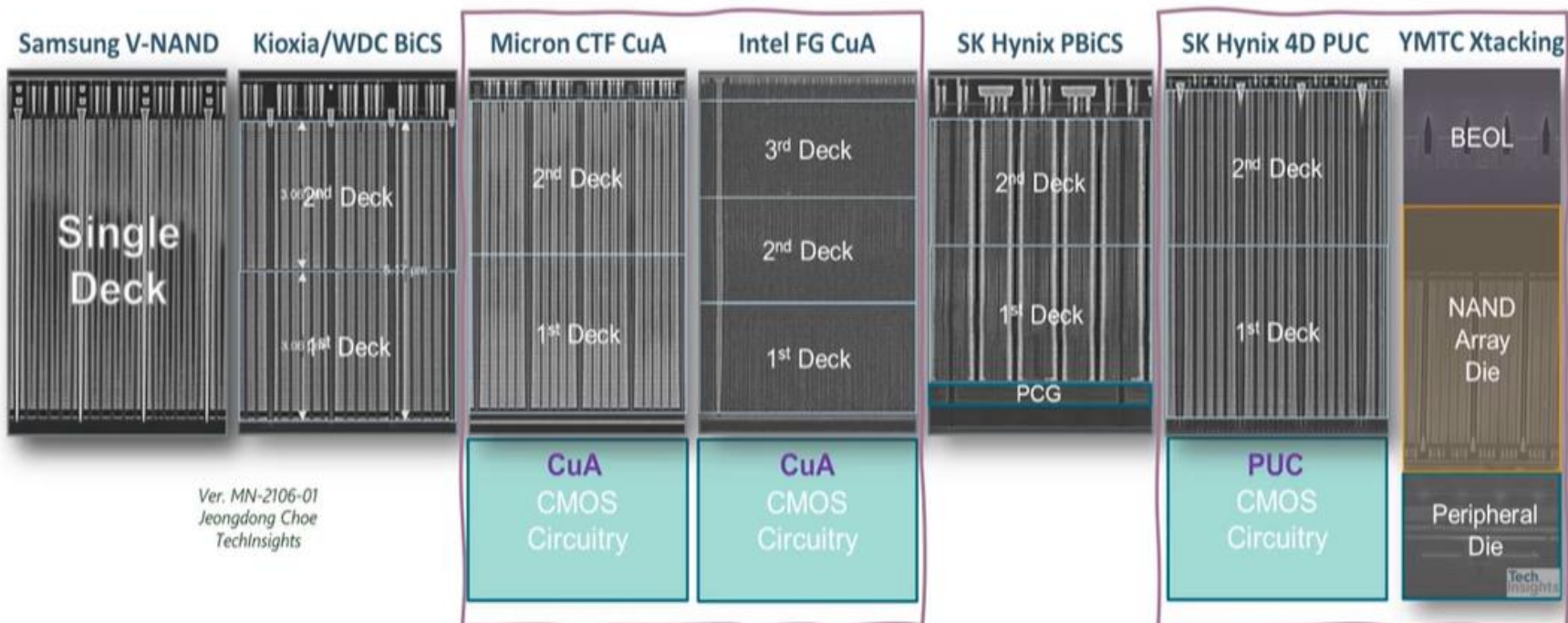
CMOS Under the Array
aka: underground parking

Small die size.
But more complex process and larger
CapEx.
More parallelism in program.

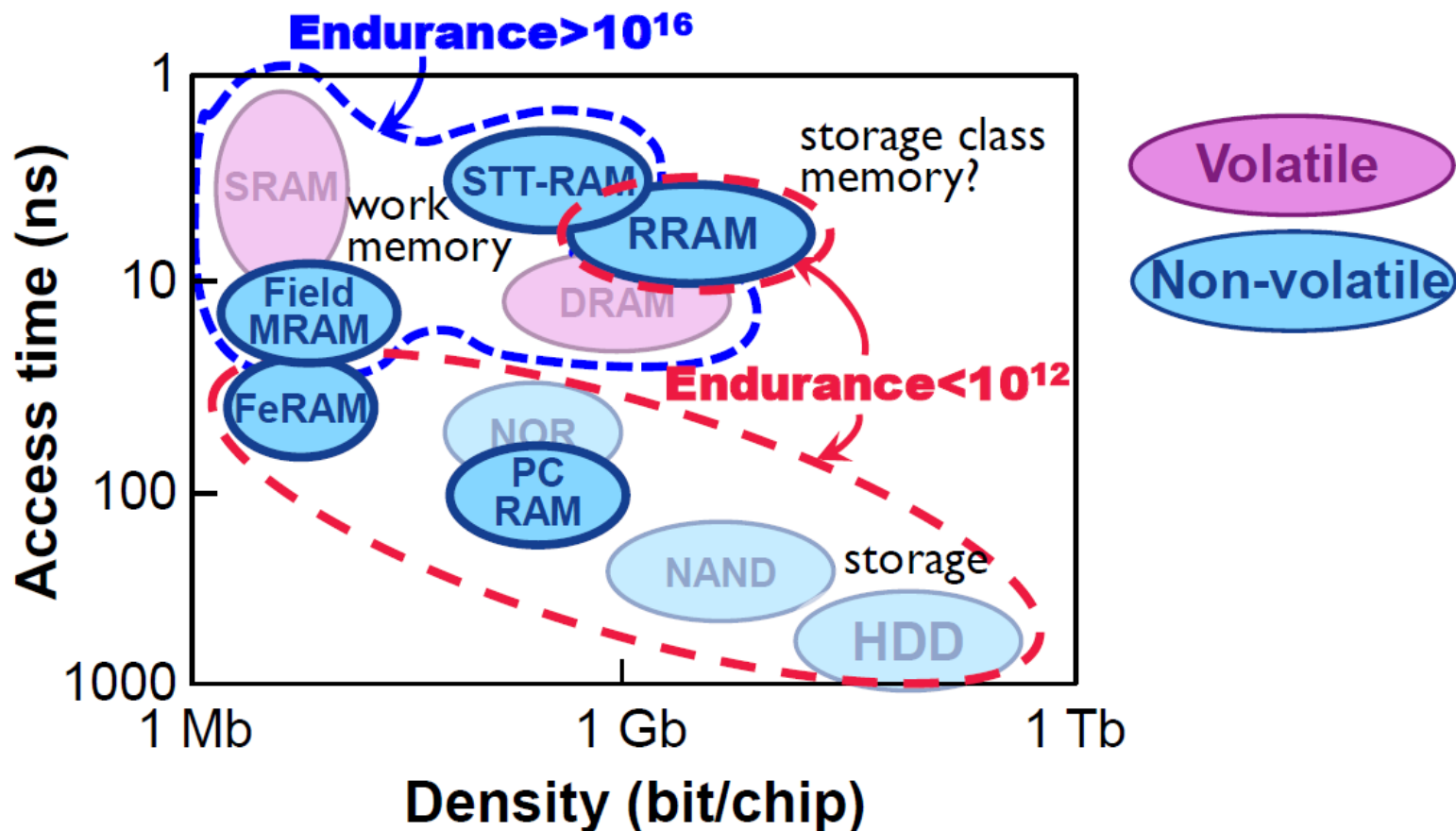


CMOS Above the Array
aka: rooftop parking

Small die size.
But more process steps, larger CapEx and
extra wafer cost.
More parallelism in program.
Can be shorter cycle time with more tools.



3. 新型存储器

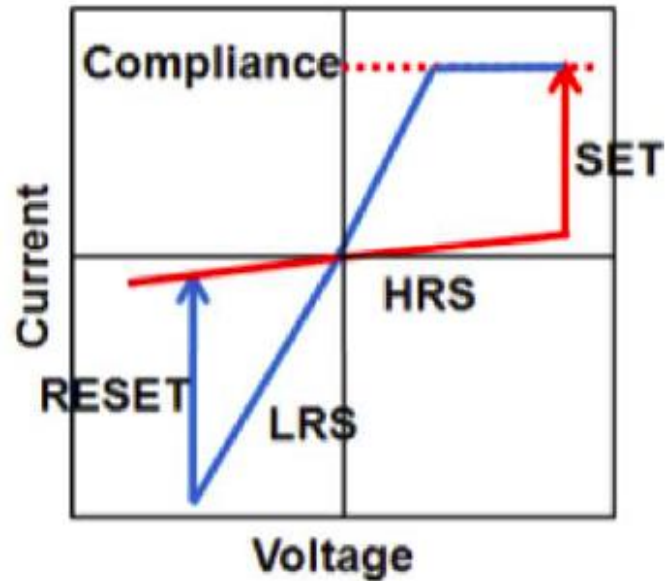
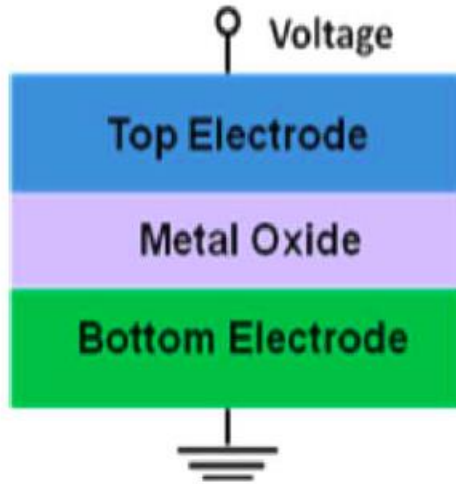


PC RAM, FeRAM, field MRAM: niche markets, embedded flash, NOR replacement

RRAM: embedded non-volatile, NAND replacement? Storage class memory

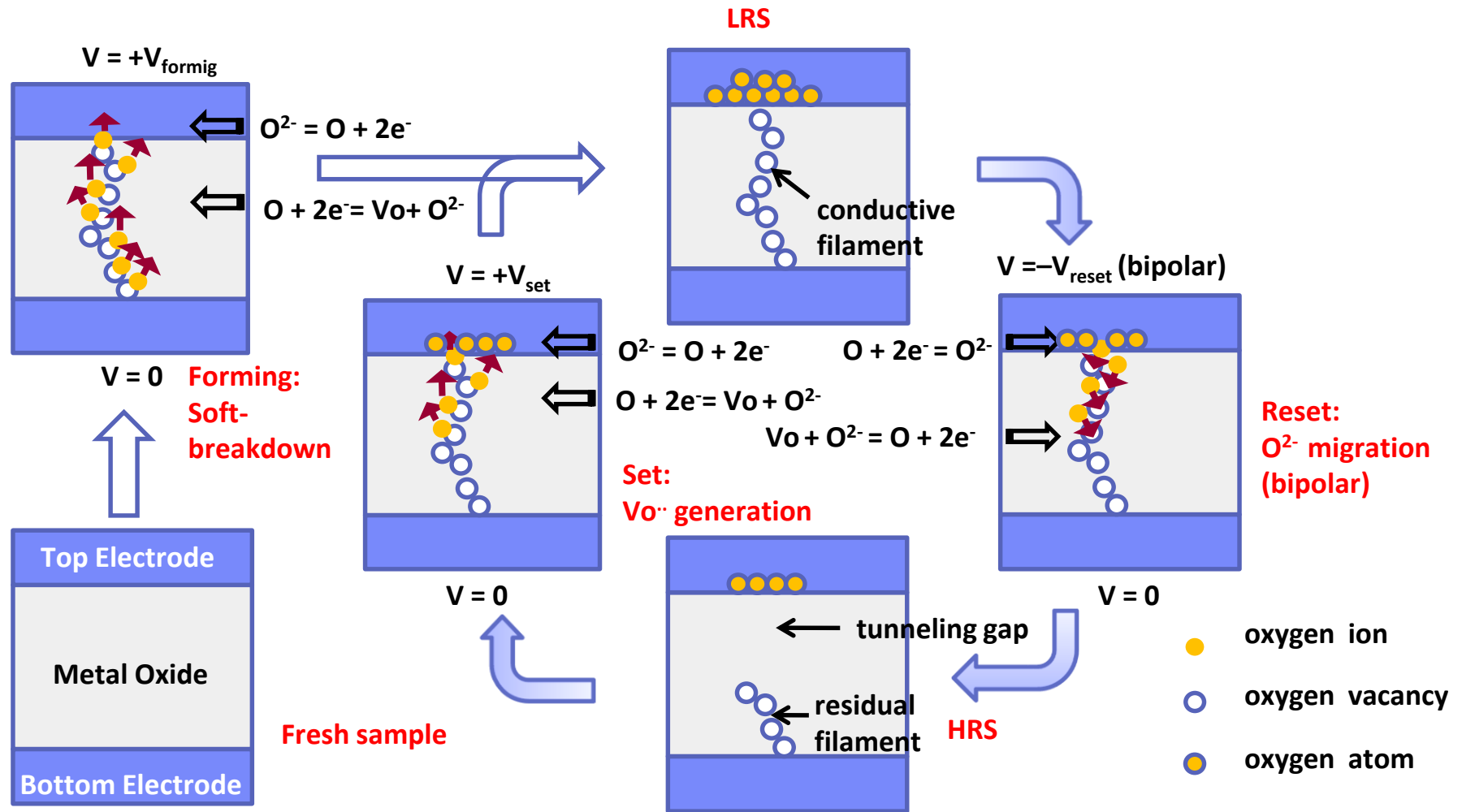
STT-RAM: DRAM replacement, embedded non-volatile, SRAM replacement?

3.1 RRAM



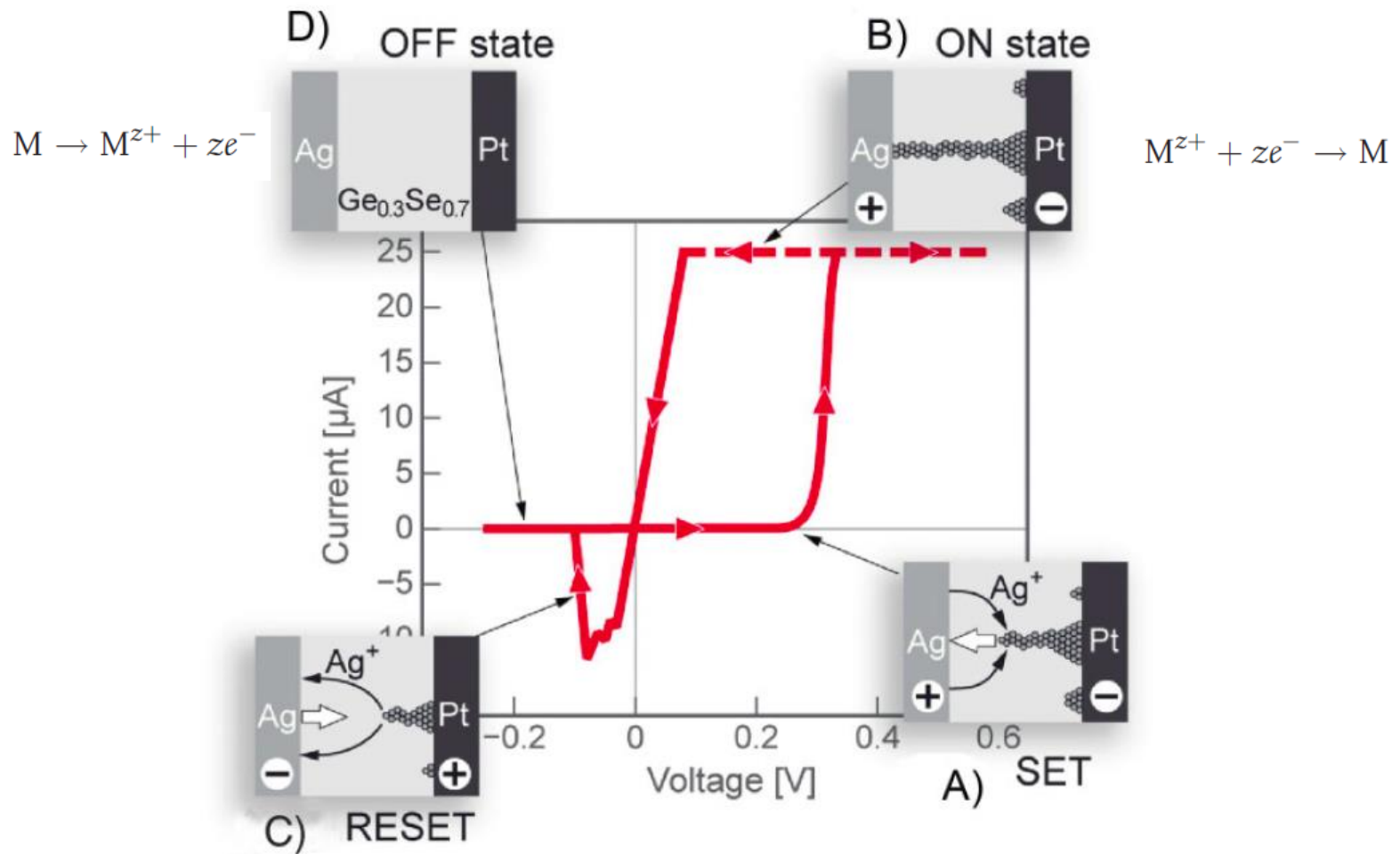
这里器件又被称为阻变存储器（**RRAM**），氧化物基存储器（**OxRAM**）

Oxide RRAM Principles



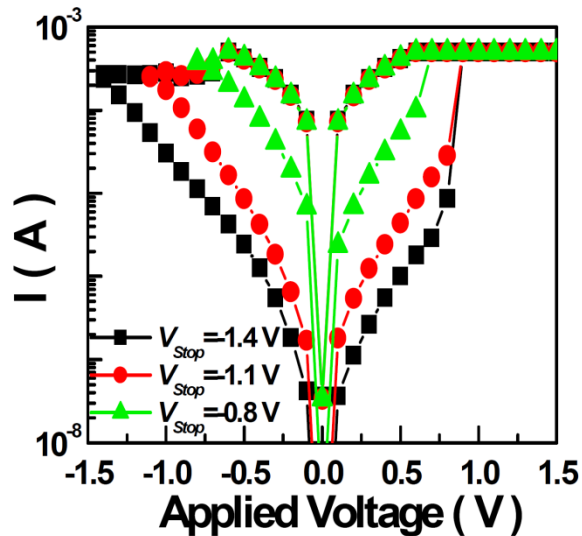
H.-S. P. Wong, H. Lee, S. Yu, et al. "Metal Oxide RRAM," Proc. IEEE, vol. 100, no. 6, pp. 1951-1970, 2012

金属通道型阻变存储器，又称导电桥型存储器 (CBRAM)

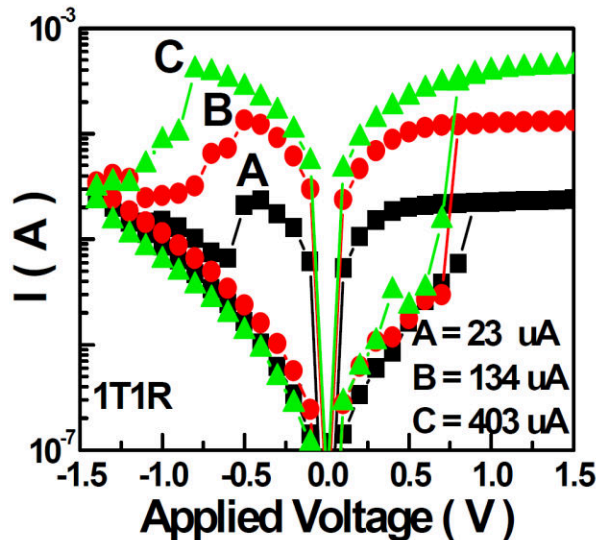


Multi-Level Operation

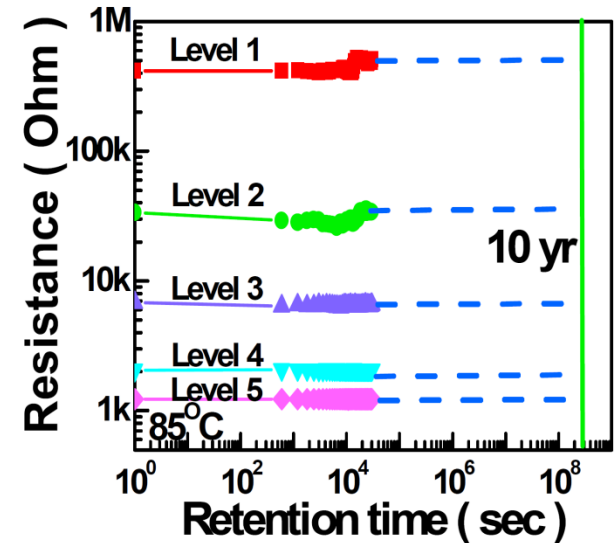
Partial Reset by reset voltage



Partial Set by set compliance current

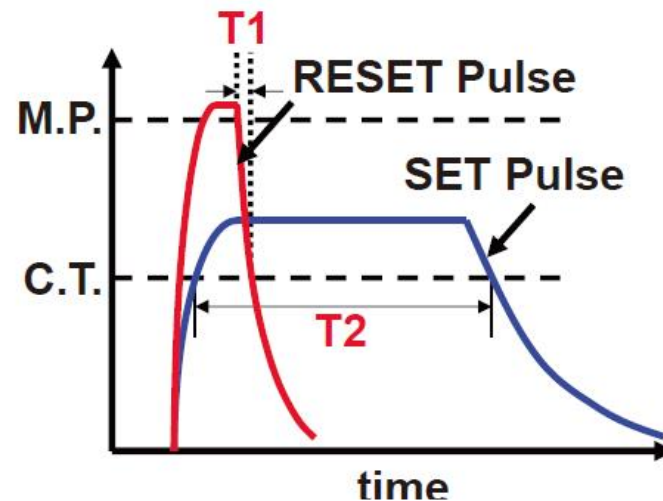
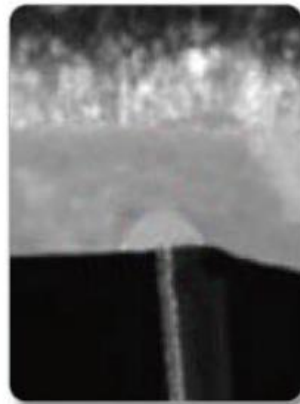
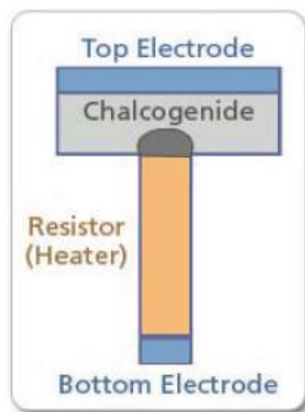


Retention of MLC

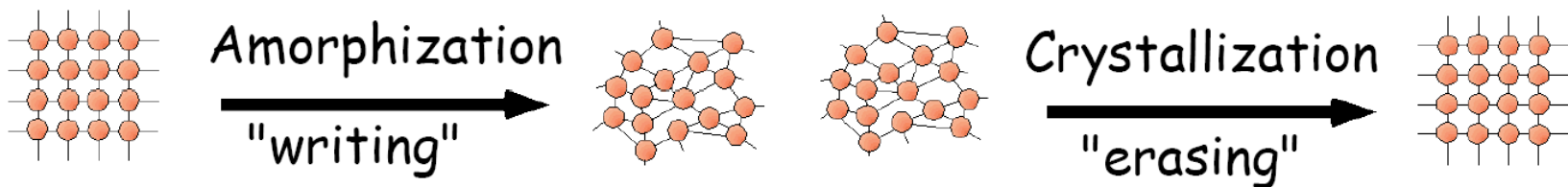
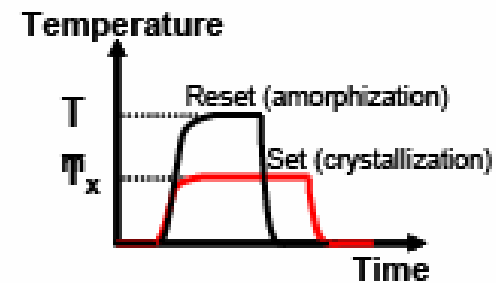
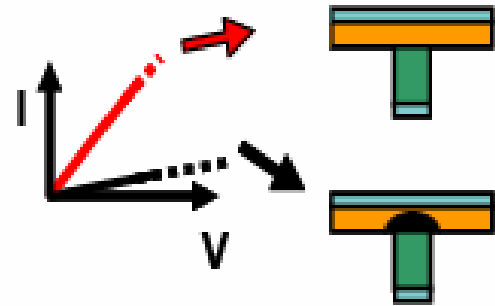


3.2 PCM

相变存储器： 英文是Phase Change Memory (PCM) 或Phase Change Random Access Memory (PRAM或PcRAM)，一般指多晶与非晶相变，表现为电压脉冲作用下的电阻变化

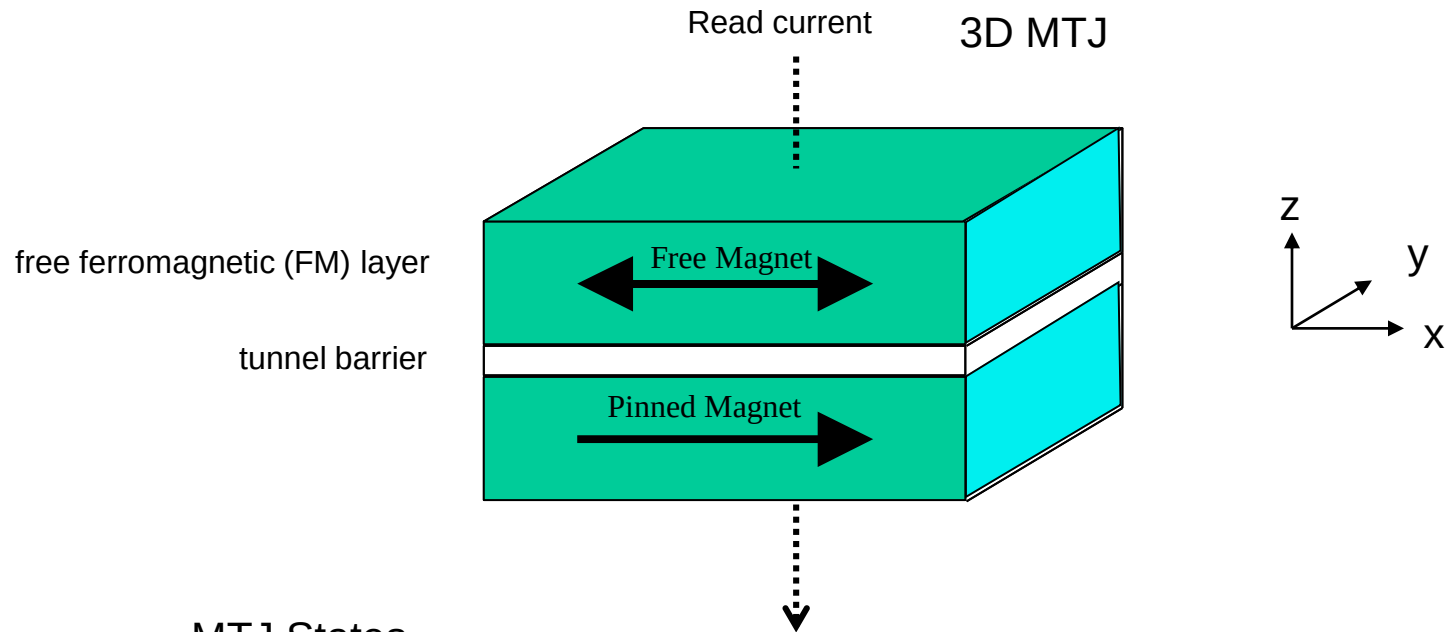


- 相变机制
 - 非晶/多晶硫族化合物合金
($\text{Ge}_2\text{Sb}_2\text{Te}_5$ -GST)
- 写机制
 - 电流焦耳效应
- 感应机制
 - GST的电阻变化

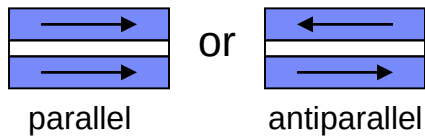


3.3 MRAM

Magnetic Tunnel Junction (MTJ)

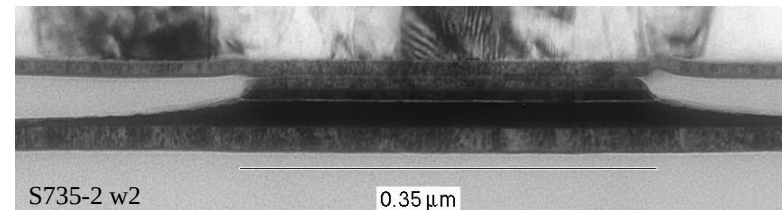


MTJ States

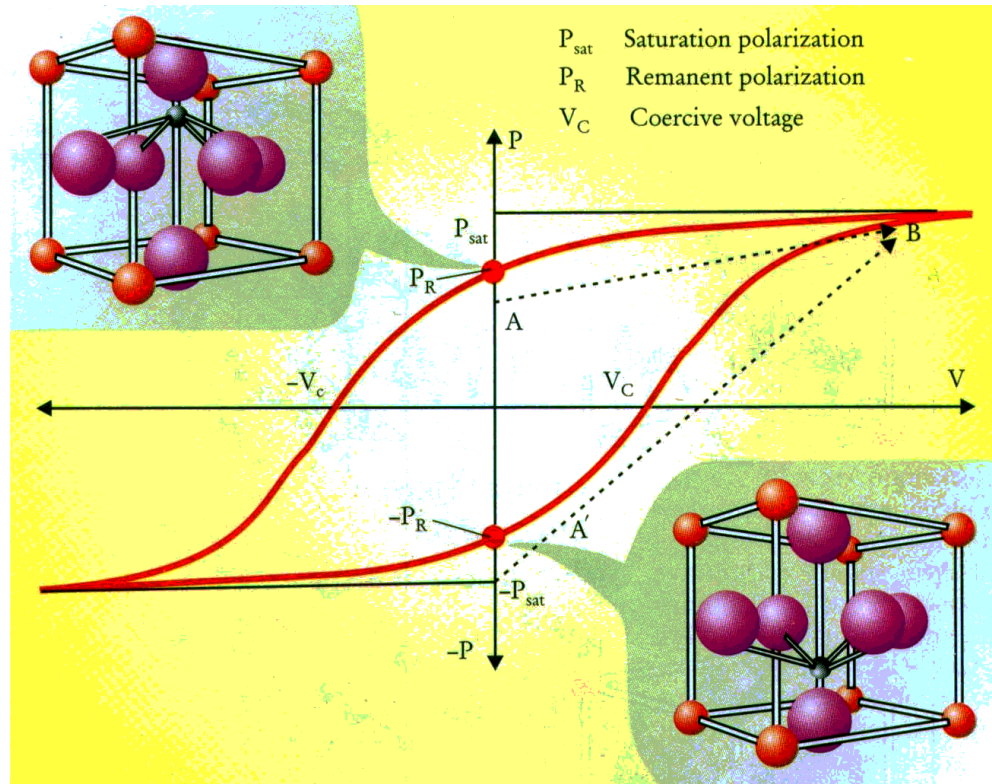


$$R_{\text{parallel}} < R_{\text{antiparallel}}$$

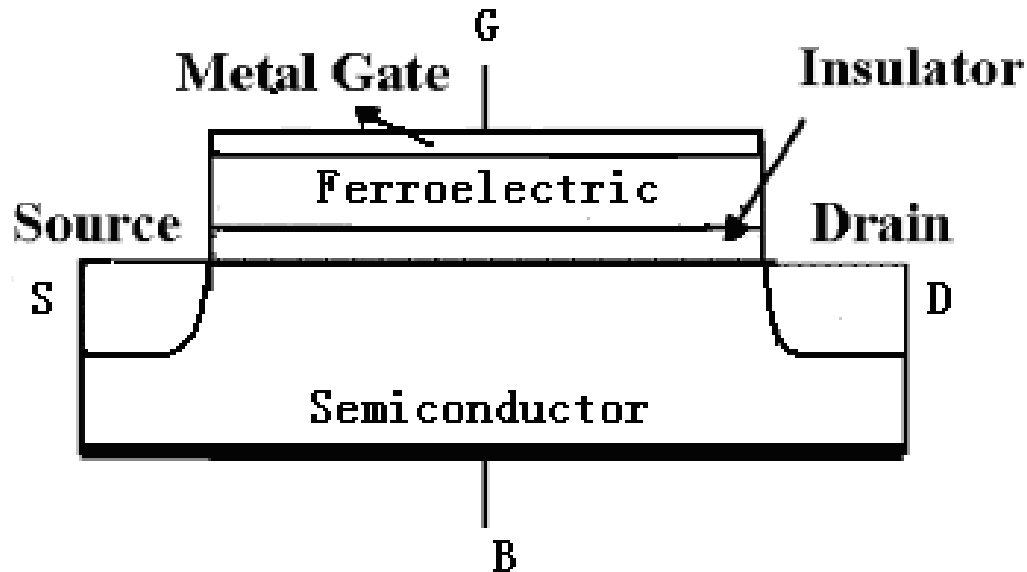
Fabricated MTJ



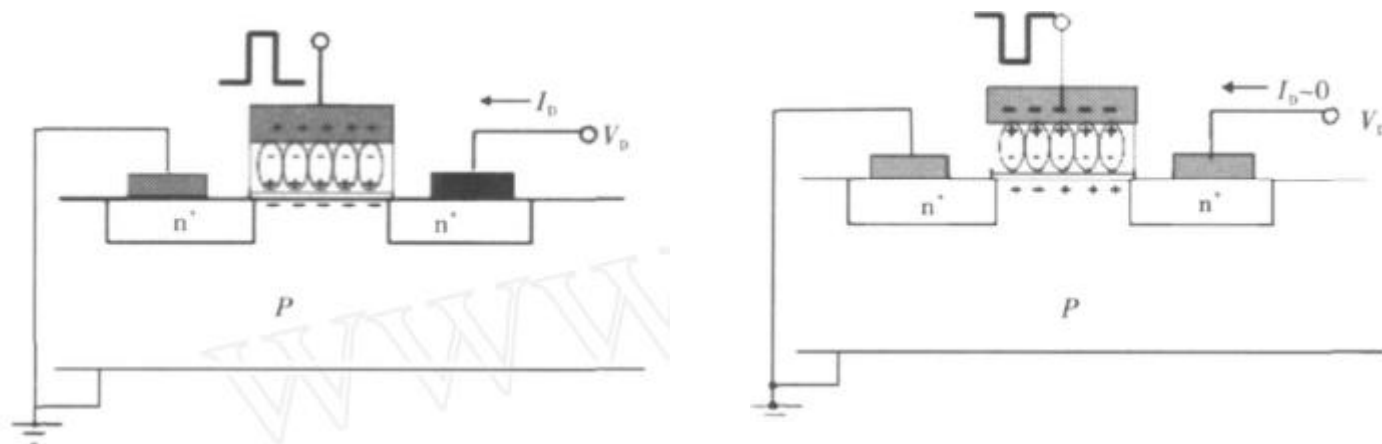
3.4 铁电存储器



铁电场效应晶体管 (FeFET)



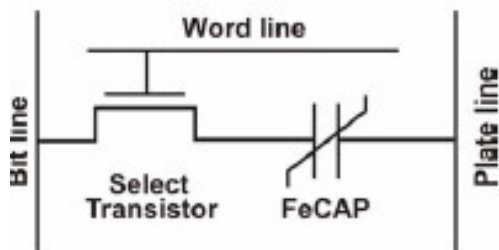
铁电场效应晶体管 (FeFET) 示意图



铁电存储器

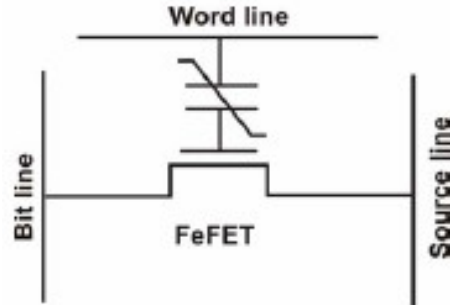
FeRAM存在主要问题：

- ✓ 存储密度
- ✓ 工艺兼容性
- ✓ 破坏性读取

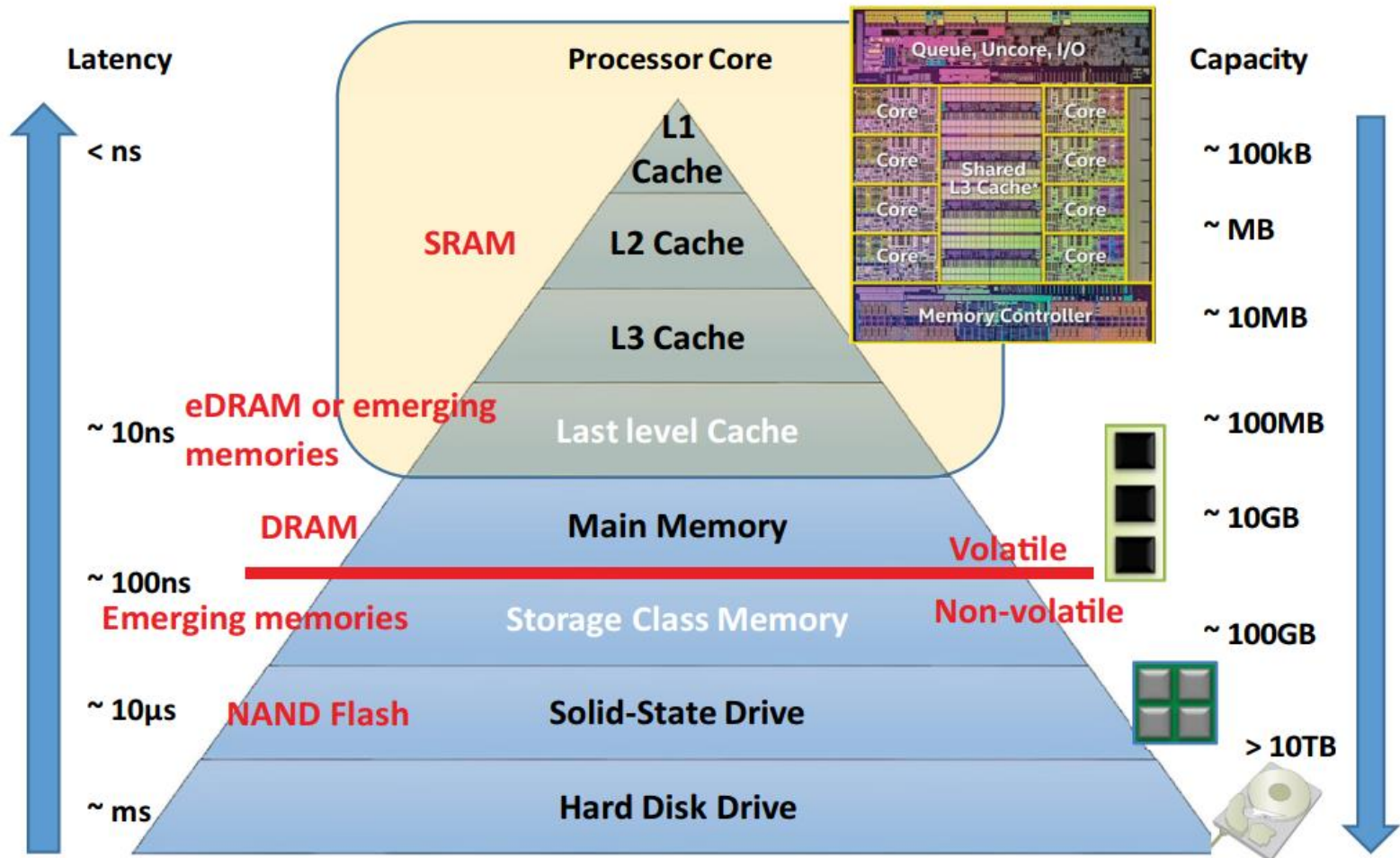


FeFET的主要问题：

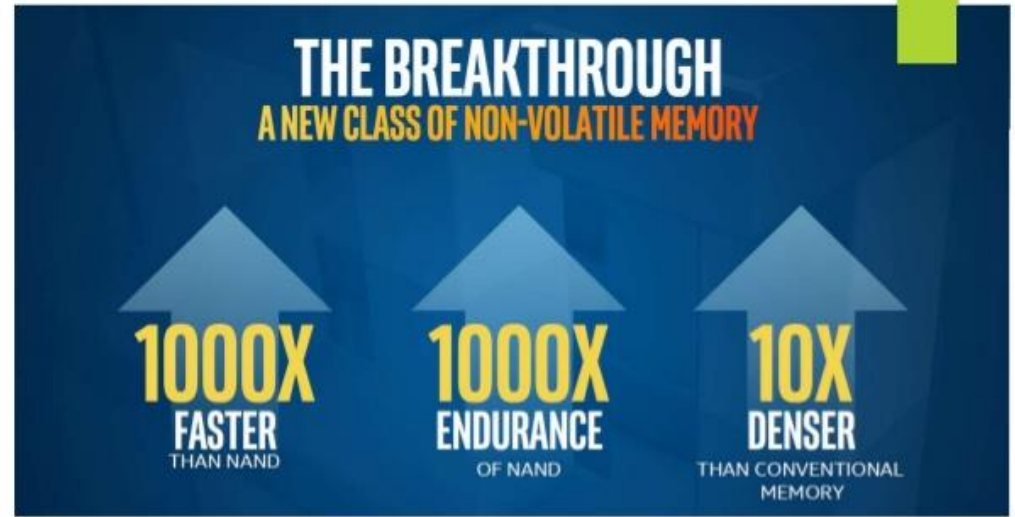
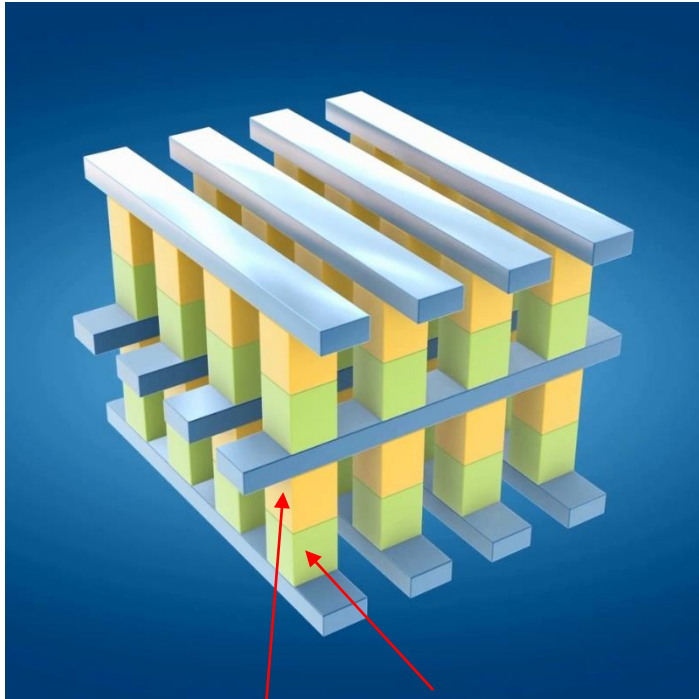
- ✓ 保持特性差
- ✓ 界面特性
- ✓ 退极化场



3.5 新型存储器系统与应用



Intel/Micron Announced 3D X-point Memory (2015)



Initial target: SCM between DRAM and NAND

Possibly PCM memory?

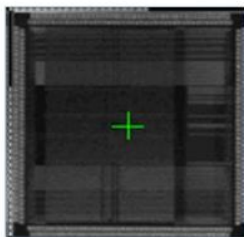
Possibly OTS threshold selector?

嵌入式存储和Cache替代

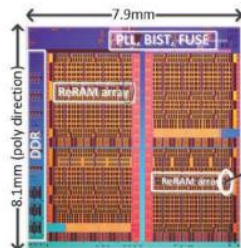
Foundry Availabilities

- Emerging memories are available from various industrial platforms (mainly for embedded NVMs, e.g., eFlash replacement).

RRAM:

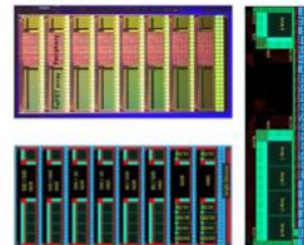


TSMC 40nm RRAM (ISSCC 18)



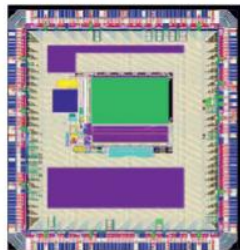
Intel 22nm RRAM (ISSCC 19)

FeFET:



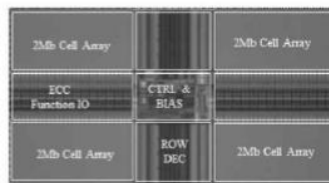
GF 28nm and 22nm FeFET (IEDM 16 & 17)

PCM:

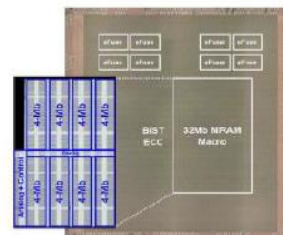


TSMC 40nm PCM (IEDM 18)

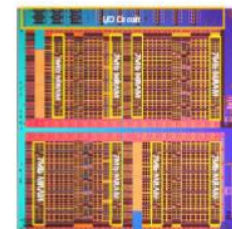
STT-MRAM:



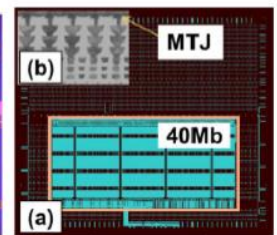
Samsung 28nm STT (IEDM 18)



TSMC 22nm STT (ISSCC 20)

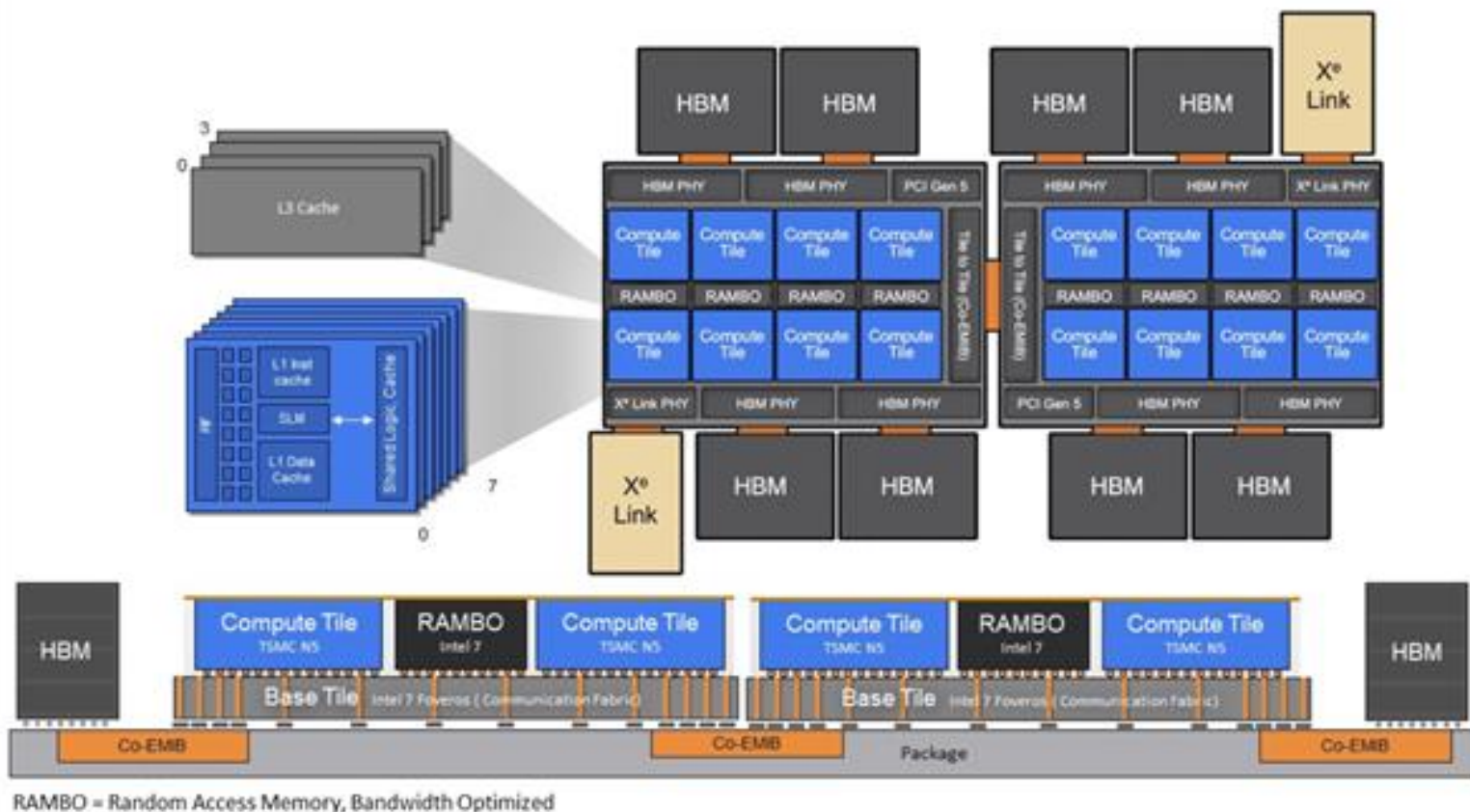


GF 22nm STT (IEDM 19)



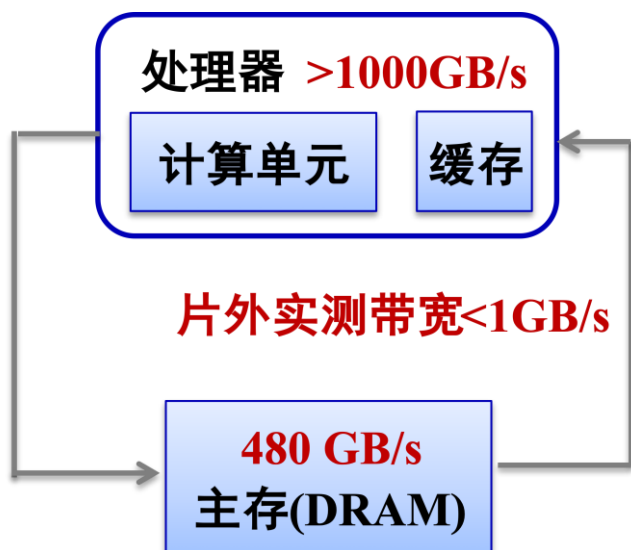
未来趋势：更多样化的存储层级

- 每个核包含专用的L1 I-Cache、L1 D-Cache，共4Mb
- 每个计算单元的多个核心共享15MB RAMBO缓存(L3 Cache);
- 芯片集成了8个HBM DRAM单元

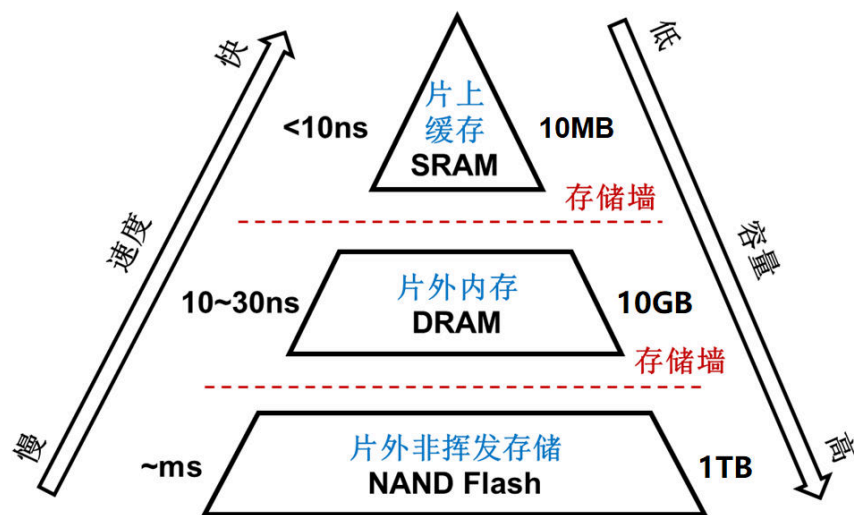


4. 存算一体

- 由于存储工艺与逻辑工艺不兼容、冯诺依曼架构延续下来的存算分离设计，产生了存储器与计算单元间大量数据搬运所造成的性能瓶颈，即“冯诺依曼瓶颈”问题
- 不同层级的存储器之间存在巨大的速度和容量鸿沟，即“存储墙”问题。

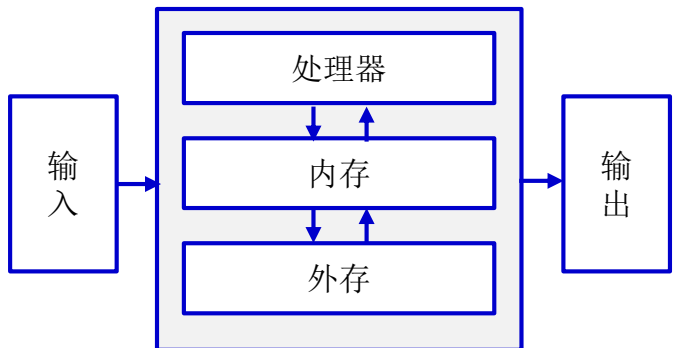


访存带宽造成算力瓶颈

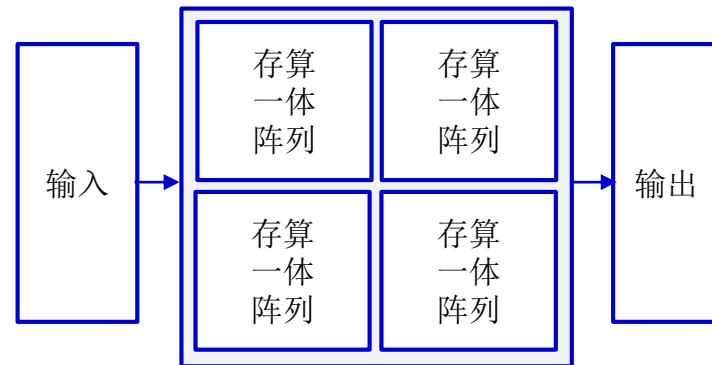


分级架构导致的存储墙瓶颈

冯诺依曼架构



存算一体架构

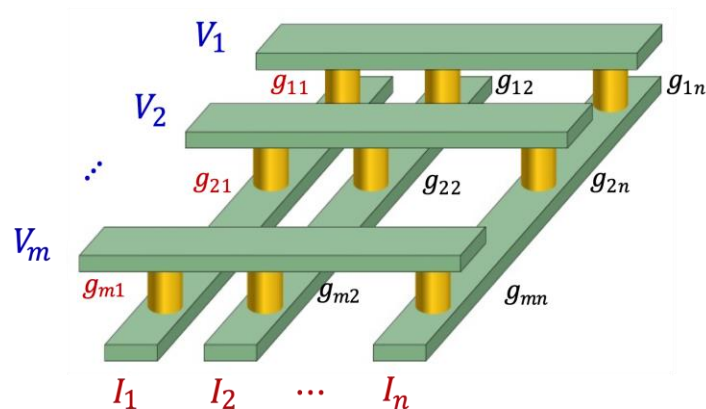


$$[x_1 \ x_2 \ \dots \ x_m] \begin{bmatrix} g_{11} & g_{12} & \dots & g_{1n} \\ g_{21} & g_{22} & \dots & g_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ g_{m1} & g_{m2} & \dots & g_{mn} \end{bmatrix}$$

向量 × 矩阵

||

$$[I_1 \ I_2 \ \dots \ I_n]$$



基于物理定律的矩阵向量乘法 —— 实现存算一体新计算范式

调研报告主题（二选一）

- 主题1：3D DRAM的最新技术
- 主题2：3D NAND的发展趋势
- 要求：文献量不少于8篇，其中精读不少于4篇，发表时间在2020年及以后，字数不少于5000